Appendices

Appendix A Forest Stewardship Plan Mapping

The attached mapping shows three individual proposed FDUs under the FSP. These areas show where all primary forest activities will take place during the term of the FSP and are described as follows:

- FDU #1 – Holberg Landscape Unit
- FDU #2 – Keogh Landscape Unit
- FDU #3 - Marble Landscape Unit

Appendix B Stocking Standards

North Island Community Forest LP STOCKING STANDARDS Even-Aged

Biogeoclimatic Zone Classification			Regeneration Guide								
			Species					Maximum Regen Delay Period	Assessment	Minimum Height at Free Growing	
			Conifer/Broadleaf		Stocking						
Zone	Subzone/Variant	Site Series	Pref. (P)	Accep. (A)	Target (well-spaced/ha)	Min (well-spaced/ha)	MITD*		Free Growing Period	Species	Min (cm)
CWH	vh1	01	Cw	Pl[6]						Cw, Yc, Pl	150
			Hw	Ba**	900	500	2.0	6	20	Hw	200
			Yc	Ss***						Fd, Ss	300
			Fd*							Ba	175
CWH	vh1	02	Pl	Hw						Pl, Hw	125
			Cw		400	200	2.0	6	20	Cw, Yc	100
			Yc								
CWH	vh1	03	Cw							Cw, Yc	100
			Hw		800	400	2.0	6	20	Hw, Pl	125
			Yc								
			Pl								
CWH	vh1	04	Ba**[47]	Ss***[35]	900	500	2.0	6	20	Ba	225
			Hw							Hw	175
			Cw							Cw	200
										Ss	400
CWH	vh1	05	Cw	Hw[2]						Cw, Yc	200
			Ba**[47]	Ss***[35]	900	500	2.0	6	20	Ba	225
			Yc							Hw	275
										Ss	400
CWH	vh1	06	Cw	Hw[2]						Cw, Yc	200
			Ba**[47]	Ss***[35]	900	500	2.0	6	20	Ba	225
			Yc							Hw	175
										Ss	400
CWH	vh1	07	Cw	Hw[2]						Cw	200
			Ba**[47]	Ss***[35]	900	500	2.0	6	20	Ba	225
										Hw	175
										Ss	400
CWH	vh1	08	Cw	Ba**[47]						Cw	200
			Ss***	Hw	900	500	2.0	6	20	Ss	400
										Ba	225
										Hw	175
CWH	vh1	09	Ss***[1]	Ba[1,47]	900	500	2.0	6	20	Ss	400
			Cw[1]							Cw	200
										Ba	225
CWH	vh1	11	Cw[1]							Cw, Yc	100
			Hw[1]		800	400	2.0	3	20	Hw, Pl	125
			Yc[1]								
			Pl[6]								
CWH	vh1	12	Pl[1]							Pl	125
			Cw		400	200	2.0	3	20	Cw, Yc	100
			Yc								
CWH	vm1	01	Hw	Ss***	900	500	2.0	6	20	Fd, Hw, Ss	300
			Fd[9,16]	Pw****						Ba	175
			Ba**[26,47]							Cw	150
			Cw							Pw	250
CWH	vm1	02	Cw	Hw	400	200	2.0	6	20	Hw	200
			Pl							Cw	100
										Pl	125
CWH	vm1	03	Cw	Pl[53]	800	400	2.0	6	20	Cw	100
			Hw							Hw, Fd	200
			Fd[9,16]							Pl	125
CWH	vm1	04	Cw	Ba**	900	500	2.0	6	20	Hw, Fd, Ss	300
			Hw	Ss***						Ba	175
			Fd[9,16]							Cw	150
CWH	vm1	05	Cw	Ss***[35]	900	500	2.0	6	20	Hw, Fd, Ss	300
			Fd[1,9,16]	Pw****						Ba	175
			Hw							Cw	150
			Ba**[47]							Pw	250
CWH	vm1	06	Ba**[26,47]	Ss***	900	500	2.0	6	20	Hw, Ss	300
			Cw							Ba	175
			Hw							Cw	150
CWH	vm1	07	Cw	Ss***[35]	900	500	2.0	6	20	Hw, Fd, Ss	400
			Hw							Ba	225
			Fd[1,9,23]							Cw	200
			Ba**[47]								

North Island Community Forest LP STOCKING STANDARDS Even-Aged

Biogeoclimatic Zone Classification			Regeneration Guide								
			Species					Maximum Regen Delay Period	Assessment	Minimum Height at Free Growing	
			Conifer/Broadleaf		Stocking						
Zone	Subzone/Variant	Site Series	Pref. (P)	Accep. (A)	Target (well-spaced/ha)	Min (well-spaced/ha)	MITD*		Free Growing Period	Species	Min (cm)
CWH	vm1	08/09	Cw Hw[2] Ba**[47]	Ss***[35]	900	500	2.0	6	20	Hw, Ss Ba Cw	400 225 200
CWH	vm1	10	Cw[1]	Ba**[1,47] Ss***[1,35]	900	500	2.0	6	20	Ss Ba Cw	400 225 200
CWH	vm1	12	Cw[1] Hw[1] Yc[1]	Pl[1]	800	400	2.0	6	20	Cw, Yc Pl Hw	100 125 200
CWH	vm1	13	Pl[1]	Cw[1]	400	200	2.0	6	20	Cw Pl	100 125
CWH	vm1	14	Cw[1]	Hw[1] Pl [53] Ss***[1,35]	800	400	2.0	6	20	Ss, Hw Pl Cw	300 200 150
CWH	vm1	01, 05, 06, 07, 08, 09, 10	Dr^		1500	1000	2.0	3	20	Dr	400
CWH	vm2	01	Cw[14] Yc[13] Hw Ba**[47] Fd[1,9,14]	Hm[13] Ss***[7,15,35]	900	500	2.0	6	20	Ss Hw Ba Fd Cw, Yc Hm	300 250 175 225 150 100
CWH	vm2	02	Cw Hw Fd[9,16] Pl Yc	Hm[13]	400	200	2.0	6	20	Hm Hw Fd Pl Cw, Yc	75 175 150 125 100
CWH	vm2	03	Cw Hw Fd[9,16] Yc	Pw****[16,31] Hm[13]	800	400	2.0	6	20	Pw Hw Fd Cw, Yc Hm	250 175 150 100 75
CWH	vm2	04	Cw Hw Yc Fd[9,16]	Ba**[47] Pw**** Ss***[35] Hm[13]	900	500	2.0	6	20	Pw Ss Ba, Fd Hw Cw, Yc Hm	250 200 150 175 100 75
CWH	vm2	05	Cw Hw Ba**[47] Yc Fd[1,9,23]	Ss***[15,35] Hm[13] Pw****	900	500	2.0	6	20	Hw, Pw Cw, Yc Ba Ss Fd Hm	250 150 175 300 400 100
CWH	vm2	06/07	Cw Hw[2] Yc Ba**[47] Fd[1,9,23]	Ss***[7,15,35] Hm[13] Pw****	900	500	2.0	6	20	Hw Ba Cw, Yc Fd Hm Pw Ss	250 175 150 400 100 250 300
CWH	vm2	08	Cw[14] Hw[2,30] Yc Ba**[47]	Ss***[30,35] Hm[13] Pw****	900	500	2.0	6	20	Ss Hw Ba Pw Cw, Yc Hm	400 350 225 250 200 100
CWH	vm2	09	Cw[1] Hw[1] Yc[1] Ba**[47]	Pl[1] Hm[13] Pw****	800	400	2.0	6	20	Hw Ba Pl Cw, Yc Pw Hm	175 150 125 100 250 75

North Island Community Forest LP STOCKING STANDARDS Even-Aged											
Biogeoclimatic Zone Classification			Regeneration Guide								
			Species					Maximum Regen Delay Period	Assessment	Minimum Height at Free Growing	
			Conifer/Broadleaf		Stocking						
Zone	Subzone/Variant	Site Series	Pref. (P)	Accep. (A)	Target (well-spaced/ha)	Min (well-spaced/ha)	MITD*		Free Growing Period	Species	Min (cm)
CWHvm2	vm2	10	Pl[1] Yc[1]	Hm	400	200	2.0	6	20	Pl Yc Hm	125 100 75
CWHvm2	vm2	11	Cw[1] Yc[1]	Hm[13,53] Hw[1] Ss***[1,15,35]	800	400	2.0	6	20	Hw Cw, Yc Hm Ss	175 100 75 200

Footnotes:

*Superscript numbers refer to footnotes from the Forest Development Plan Guidance to Stocking Standards (updated to March 2019), on the following page.

MITD = Minimum inter-tree distance is reduced to 1.6 metres on those sites where there are limited plantable spots. 1. Raised microsites on wet ground, 2. Sites with forest floor over bedrock where plantable spots are limited, and 3. Heavy accumulations of woody debris.

*Fd is restricted to southerly aspects.

** Ba is restricted to nutrient medium or rich microsites with elevated microsites preferred.

***Ss - Sitka spruce has a risk of weevil damage. If planted, will be restricted to weevil resistant stock and will make up no more than 10 - 30% of the well-spaced stems per hectare

****Pw is at risk for White Pine Blister Rust, only resistant stock will be used, or the lower branches of trees are pruned to 3 metres within the free growing window.

^ Dr Management

1. MSS reduced using approved variation to 800 sph on 20% of the area (ha) managed for hardwoods due to site limiting factors as a separate Standards Unit.
2. CWHvm1 01, alder is limited to 3-4/C on SMR/SNR table
3. CWHvm1 06, alder is limited to 5-6/C on SMR/SNR table

Footnotes below apply to FSP Stocking Standards Even-Aged

Conifer Tree Species

"Ba" means amabilis fir;
"Bg" means grand fir;
"Bl" means subalpine fir;
"Bp" means noble fir;
"Cw" means western red cedar;
"Fd" means Douglas-fir;
"Hm" means mountain hemlock;
"Hw" means western hemlock;
"Lt" means tamarack;
"Lw" means western larch;
"Pa" means whitebark pine;
"Pl" means lodgepole pine;
"Pw" means white pine;
"Py" means ponderosa pine;
"Sb" means black spruce;
"Se" means Engelmann spruce;
"Ss" means Sitka spruce;
"Sw" means white spruce;
"Sx" means hybrid spruce or interior spruce;
"Sxs" means hybrid Sitka spruce;
"Sxw" means hybrid white spruce;
"Yc" means yellow cedar.

Broadleaf Tree Species

"Acb" means balsam poplar;
"Act" means black cottonwood;
"At" means trembling aspen;
"Dr" means red alder;
"Ep" means common paper birch;
"Mb" means bigleaf maple;
"Qg" means garry oak;
"Ra" means arbutus;

"Biogeoclimatic unit" or "BGC classification" means the zone, subzone, variant and site series described in the most recent field guide published by the Ministry of Forests for the identiication and interpretation of ecosystems, as applicable to a harvested area.

"**MIN** or "**Min**" means minimum.

Footnote #	Footnote
1	suitable on elevated microsites
2	retired July 2017
3	suitable on coarse-textured soils
4	suitablemedium-textured soils
5	footnote retired
6	suitable on nutrient-very-poor sites
7	suitable on nutrient-medium sites
8	suitable on steep slopes
9	suitable on warm aspects
10	suitable on cool aspects
11	suitable on crest slope positions
12	suitable on cold air drainage sites
13	suitable at upper elevations
14	suitable at lower elevations
15	suitable in the northern portion of biogeoclimatic unit
16	suitable in the southern portion of biogeoclimatic unit
17	suitable in the western portion of biogeoclimatic unit
18	suitable in the eastern portion of biogeoclimatic unit
19	retired July 2017
20	retired July 2017
21	retired July 2017
22	suitable in the southern Gardner Canal-Kitlope area
23	retired July 2017
24	suitable in wetter portion of biogeoclimatic unit
25	retired July 2017
26	suitable minor species on nutrient poor sites
27	partial high-canopy shade required for succesful establishment
28	limited by moisture deficit
29	risk of heavy browsing by moose
30	retired November 2010
31	must use of blister rust resistant stock. See BC Journal of Ecosystems and Management 10(1): 97-100 for supplementary information.
32	limited by growing-season frosts
33	footnote retired and replaced with footnote 'a'
34	risk of snow damage
35	use resistant stock to mitigate risk of spruce weevil damage - See Ss Weevil Decision Tool: http://pubs.cif-ifc.org/doi/abs/10.5558/tfc2013-042
36	retired July 2017
37	retired November 2010
38	footnote retired
39	retired July 2017
40	risk of redheart damage in areas subject to cold winter outflow winds
41	limited by poorly drained soils
42	suitable on sites with a fresh soil moisture regimes
43	retired July 2017
44	suitable in areas of the subzone variant with relatively strong maritime influence
45	suitable in areas of the subzone variant with relatively strong continental influence
46	use resistant seedlot south of the Dean Channel
47	risk of balsam wooly adelgid within quarantine area see http://www2.gov.bc.ca/gov/content/industry/agriculture-seafood/animals-and-crops/plant-health/insects-and-plant-diseases/nursery-and-ornamentals/balsam-woolly-adelgid
48	risk of browsing by deer
49	retired November 2010
50	restricted to sites where the species occurs as a major species in a pre-harvest, natural stand
51	retired July 2017
52	suitable on sheltered microsites with deep soil
53	minor component
54	retired July 2017
55	retired July 2017

#	Broadleaf Management Constraints
a	productive, reliable, and feasible regeneration option
b	limited in productivity, reliability and/or feasibility

#	Localized Footnotes
56	retired July 2017
57	retired November 2010
58	**South Area** - Fd limited to a max 50% of preferred and acceptable well-spaced stems in the IDFmw and all subzones of the ICH due to root rot. See Root Rot Handbook for management issues (FLNRORD 2018).

Appendix C Vancouver Island Land Use Plan Higher Level Plan Order

Order Establishing Resource Management Zones and Resource Management Zone Objectives within the area covered by the Vancouver Island Land Use Plan, pursuant to sections 3(1) and 3(2), as well as section 9.1 of the Forest Practices Code of British Columbia Act (the Act)

I. Pursuant to section 3(1) of the Act, the following zones, as presented on Map 1 (attached), are Resource Management Zones (RMZ):

A. Special Management Zones (SMZ) 1 through 14 and 17 through 22;

B. Resource Management Zones 4, 5, 6, 8, 10, 11, 15, 18, 19, 21, 23, 24, 27, 28, 30, 36, 38, 40, 42, 43, 44, and 47; these RMZ are also referred to as Enhanced Forestry Zones (EFZ);

C. Resource Management Zones 7 and 14; these two RMZ are also referred to as General Management Zones (GMZ).

II. Pursuant to section 3(2) of the Act, the following provisions are Resource Management Zone objectives:

A. *for Special Management Zones 1 through 14 and 17 through 22:*

1. Sustain forest ecosystem structure and function in SMZs, by:
- (a) creating or maintaining stand structures and forest attributes associated with mature[1] and old[2] forests, subject to the following:
  - i. the target for mature seral forest should range between one quarter to one third of the forested area of each SMZ[3]; and
  - ii. in SMZs where the area of mature forest is currently less than the mature target range referred to in (i) above, the target amount of mature forest must be in place within 50 years;
- (b) retaining, within cutblocks[4], structural forest attributes and elements with important biodiversity functions[5]; and
- (c) applying a variety of silvicultural systems, patch sizes and patch shapes across the zone, subject to a maximum cutblock size of 5 ha if clearcut, clearcut with reserves or seed tree silvicultural systems are applied, and 40 ha if shelterwood, selection or retention silvicultural systems are applied[6].

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[1] The mature seral forest is defined as generally 80 to 120 years old or older, depending on species and site conditions. The structure of mature seral forests generally includes canopies that vary vertically or horizontally, or both. The age and structure of the mature seral stage will vary significantly by forest type and from one biogeoclimatic zone to another.

[2] The old seral forest is defined as generally greater than 250 years old, containing live and dead (downed and standing) trees of various sizes, including large diameter trees, and of various tree species, including broad-leaved trees. The structure of old seral forest varies significantly by forest type and from one biogeoclimatic zone to another.

[3] Mature seral targets will be established through landscape unit planning. See transition provisions under III.

[4] Within cutblocks: generally means non-contiguous with cutblock boundaries.

[5] This includes, but is not limited to snags, wildlife trees, downed logs.

[6] Maximum cutblock sizes refer to net area to be reforested.

2. Despite subsection 1(c) above, cutblocks larger than 5 or 40 ha, as the case may be, may be approved if harvesting is being carried out to recover timber that was damaged by fire, insects, wind or other similar events and wherever possible, the cutblock incorporates structural characteristics of natural disturbances.

3. Pursuant to section 2(1) of the Operational Planning Regulation (OPR)[7], the approval of both the district manager, Ministry of Forests and the designated environment official, Ministry of Environment, Lands and Parks is required for all forest development plans, or parts of forest development plans that relate to areas within the following SMZs: 1, 3, 4, 6, 8, 9, 10, 11, 13, 17, 19, 20 and 21.

B. ***for Special Management Zones 8, and 13, and parts of Special Management Zones 1, 3 and 11, which are located within landscape units with higher biodiversity emphasis, as shown on Map 2:***

4. Maintain late-successional habitat elements and attributes of biodiversity[8] in forested ecosystems with emphasis on regionally rare and underrepresented ecosystems, by retaining old seral forest at the site series/surrogate level of representation[9].

5. Retain late-successional habitat elements and attributes of biodiversity in patches of variable size.

C. ***for the following Special Management Zones with primary visual resource values: 1, 2, 3, 5, 6, 7, 10, 11, 12, 13, 14, 18, 19 and 22, as shown on Map 3:***

6. Maintain the visual quality of known scenic areas in accordance with the recommended visual quality classes in the visual landscape inventory, until the district manager establishes visual quality objectives for the areas.

D. ***for all Enhanced Forestry Zones, as shown on Map 1, save and except the parts of those zones which are designated as community watersheds as defined in section 41(8) of the Act:***

7. To increase the short-term availability of timber,

(a) a cutblock may be larger than 40 ha pursuant to section 11(2)(a) of the OPR; and
(b) pursuant to section 68(4) of the OPR, a cutblock is greened-up if it is adequately stocked and the average height of those trees that are
(i) the tallest tree in each 0.01 ha plot included in a representative sample, and
(ii) a commercially valuable species or other species acceptable to the district manager
is at least 1.3 meters;

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[7] BC Reg. 107/98 O.C. 426/98 - Effective: June 15, 1998.

[8] This includes, but is not limited to: large diameter (> 60 cm) live, decaying and dead standing trees (providing nest and cavity sites); downed wood, including large diameter pieces (50 to 150 cm); deciduous broad-leaved trees, both in riparian and upland areas.

[9] The level of representation of old seral forest will be applied through landscape unit planning.

unless the district manager determines that a cutblock referred to under (a) or (b) would significantly impact specific hydrological, wildlife, biodiversity, scenic or recreation values.

8. Avoid or mitigate adverse hydrological impacts, which may result from the practices referred to in objective 7, in watersheds with significant watershed sensitivity and significant fisheries values, as determined by the district manager and designated environment official.

9. When proposing the species composition for the purposes of OPR section 39 (3) (o), a person may, pursuant to OPR section 41, select a single species that is ecologically suited to the area, if a mix of species was present on the area before the timber was harvested.

9.1 The area that may be subject to selection of a single species pursuant to objective 9 is limited to no more than 20 per cent of the forested area of any variant within a given EFZ.

E. *for Resource Management Zones 7 and 11:*

10. To avoid severe social and economic consequences, as determined by the district manager and the designated environment official, the full target of 13 per cent for old growth retention in CWHvm1 may be reduced by up to one third provided that ecologically suitable second growth forest is identified to recruit the shortfall[10].

F. *for Resource Management Zone 42:*

11. Retain old seral forest in CWHvm1 in accordance with the full old seral target of 13 per cent for the variant.

11.1 Despite objective 11, up to one third of the old seral target may be recruited from second growth provided that

(a) such recruitment is necessary to avoid severe social and economic consequences;
(b) such recruitment will not impact the ability to conserve suitable habitat of identified wildlife in accordance with the Identified Wildlife Management Strategy[11]; and
(c) ecologically suitable second growth forest is identified to recruit the shortfall.[12]

G. *for Special Management Zone 10:*

12. Retain or recruit old growth forest in CWHxm2 in accordance with the full old seral target of 9 per cent for the variant.

H. *for Resource Management Zone 10*

13. Retain old seral forest in CWHxm2 in accordance with the full old seral target of 9 per cent for the variant.

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[10] The targets for retention or recruitment of old growth forests will be achieved through the establishment of old growth management areas as part of landscape unit planning.

[11] See "Managing Identified Wildlife: Procedures and Measures", Volume 1, February 1999.

[12] The targets for retention or recruitment of old growth forests will be achieved through the establishment of old growth management areas as part of landscape unit planning.

13.1 Despite objective 13, up to one third of the old seral target may be recruited from second growth provided that

(a) such recruitment is necessary to avoid severe social and economic consequences;

(b) such recruitment will not impact the ability to conserve suitable habitat of identified wildlife in accordance with the Identified Wildlife Management Strategy[13]; and

(c) ecologically suitable second growth forest is identified to recruit the shortfall.[14]

I. *for Resource Management Zone 30:*

14. Retain all remaining old growth forest in CWHxm2 until landscape unit objectives for old growth retention or recruitment have been established in accordance with the full old seral target of 9 per cent for the variant.

J. *for Resource Management Zones 8, 14, 28 and 43:*

15. Retain old growth forests to meet old seral targets[15] and marbled murrelet habitat requirements[16] in the non-contributing[17] land base to the fullest extent possible.

16. Beyond retention in the non-contributing land base, retain old forests in the timber harvesting land base, up to the full target amount, if the district manager and the designated environment official determine that such retention is required to maintain critical marbled murrelet habitat[18].

III. Transition

17. Pursuant to section 9.1 of the Act, the following objectives will not be implemented in an area until landscape units and objectives have been established for the area, in accordance with section 4 of the Act:

Objectives 1(a); 4; 5; 10; 11; 11.1; 12; 13; 13.1; 15; and 16.

18. In the event that landscape units and objectives are not established in an area within 2 years of the date that this order takes effect, the objectives referred to in paragraph 17 will be implemented in the area.

IV. Filing the Order

This order will be filed with the regional manager of the Vancouver Forest Region and will take effect on December 1, 2000.

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[13] See "Managing Identified Wildlife: Procedures and Measures", Volume 1, February 1999.

[14] The targets for retention or recruitment of old growth forests will be achieved through the establishment of old growth management areas as part of landscape unit planning.

[15] See "Landscape Unit Planning Guide", March 1999.

[16] See "Managing Identified Wildlife: Procedures and Measures", Volume 1, February 1999.

[17] Non-contributing: the crown forested land base that does not contribute to the annual allowable cut, but does contribute to biodiversity objectives and targets.

[18] Retention or recruitment of old growth forests will be achieved through the establishment of old growth management areas as part of landscape unit planning.

BRITISH COLUMBIA

Vancouver Island Land Use Plan
Higher Level Plan Order

The Honourable Jim Doyle
Minister of Forests

Date 24/oct/00

The Honourable Joan Sawicki
Minister of Environment, Lands and Parks

Date 19/10/00

The Honourable Dan Miller
Minister of Energy and Mines

Date 19/10/00

Appendix D Non-Spatial Old-Growth Order

Preamble

The preamble is intended to provide context and it does not form part of the Order.

The intent of this Order, which establishes provincial non-spatial old growth objectives, is to identify the amount of old forest that will be maintained to address biodiversity values across the province. This will assist in clarifying the amount of area available for timber harvesting.

In ensuring that their plans are consistent with the objectives of this Order, licensees should consider the Implementation Policy developed by the Ministry of Sustainable Resource Management to support the Order.

This order supplements, but does not replace, section 9 of the Forest Planning and Practice Regulation.

Order Establishing Provincial Non-Spatial Old Growth Objectives

Order

I. Pursuant to section 4(1) of the *Forest Practices Code of British Columbia Act* (Act), the landscape units listed in Appendix 1, Table 1 are established as landscape units.

II. Pursuant to section 4(2) of the Act, part A of this Order establishes landscape unit objectives for the landscape units listed in Appendix 1, Table 1.

III. This Order does not apply to land contained within a woodlot licence, an area contained in a community forest agreement that is less than 600 hectares, or to areas defined as the 'trust area' as per the *Islands Trust Act* with the exception of Gambier and Anvil Island.

A. <u>Biodiversity emphasis and old growth objectives</u>

1. Biodiversity emphasis for landscape units

For the purpose of implementing objective 2 below, biodiversity emphasis is assigned as listed in Appendix 1, Table 1.

2. Old growth objectives

To contribute to the conservation of biodiversity, licensees[1] must maintain old forest by biogeoclimatic variant[2] within each landscape unit according to the age of old forest and the percentage of old forest retention that is specified in Tables 1 through 4 and the assignment of Natural Disturbance Types outlined in Appendix 3, and subject to provisions 5 through 9 below.

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[1] In this Order, "licensee" refers to a party required to prepare a forest development plan under the *Forest Practices Code of British Columbia Act* or a forest stewardship plan under the *Forest and Range Practices Act* with the exception of any agreement holder mentioned in III above.

[2] As approved by the Regional Director of MSRM, see the attached Implementation Policy.

Table 1. Natural Disturbance Type One

Biogeoclimatic Zone	Age of Old Forest	Percent Old Forest Retention in Low Biodiversity Emphasis	Percent Old Forest Retention in Intermediate Biodiversity Emphasis	Percent Old Forest Retention in High Biodiversity Emphasis
CWH[a]	>250yrs	>13	>13	>19
ICH	>250yrs	>13	>13	>19
ESSF	>250yrs	>19	>19	>28
MH	>250yrs	>19	>19	>28

a Some portions of the CWH have a much more frequent disturbance history due to extensive windthrow. Those portions of the CWHvm1, CHWvm2, CWHvh1, and CWHvh2 where wind event occur, should be considered to fall under NDT3.

Table 2. Natural Disturbance Type Two

Biogeoclimatic Zone	Age of Old Forest	Percent Old Forest Retention in Low Biodiversity Emphasis	Percent Old Forest Retention in Intermediate Biodiversity Emphasis	Percent Old Forest Retention in High Biodiversity Emphasis
CWH	>250yrs	>9	>9	>13
CDF	>250yrs	>9	>9	>13
ICH	>250yrs	>9	>9	>13
SBS	>250yrs	>9	>9	>13
ESSF	>250yrs	>9	>9	>13
SWB	>250yrs	>9	>9	>13

Table 3. Natural Disturbance Type Three

Biogeoclimatic Zone	Age of Old Forest	Percent Old Forest Retention in Low Biodiversity Emphasis	Percent Old Forest Retention in Intermediate Biodiversity Emphasis	Percent Old Forest Retention in High Biodiversity Emphasis
BWBS[b]	>100yrs	>13	>13	>19
SBPS	>140yrs	>7	>7	>10
BWBS[c]	>140yrs	>11	>11	>16
SBS	>140yrs	>11	>11	>16
MS	>140yrs	>14	>14	>21
ESSF	>140yrs	>14	>14	>21
ICH	>140yrs	>14	>14	>21
CWH[d]	>140yrs	>11	>11	>16

b BWBS with deciduous prominent

c BWBS with coniferous prominent

d CWH subject to regular extensive wind throw disturbance. See notes under NDT 1 table for a listing of variants.

Table 4. Natural Disturbance Type Four

Biogeoclimatic Zone	Age of Old Forest	Percent Old Forest Retention in Low Biodiversity Emphasis	Percent Old Forest Retention in Intermediate Biodiversity Emphasis	Percent Old Forest Retention in High Biodiversity Emphasis
ICH	>250 yrs	>13	>13	>19
IDF	>250 yrs	>13	>13	>19
PP	>250 yrs	>13	>13	>19

3. Old growth objectives for the Okanagan area

To contribute to the conservation of biodiversity in the Okanagan area[3] licensees must maintain old forest by biogeoclimatic variant within each landscape unit to the amount specified in Appendix 2, Table 1 (Okanagan Old Forest Retention Table), subject to provisions 6-8 below.

4. Old growth objectives for the Merritt area

To contribute to the conservation of biodiversity in the Merritt area[4] licensees must maintain old forest by biogeoclimatic variant within each landscape unit to the amount specified in Appendix 2, Table 2 (Merritt Old Forest Retention Table), subject to provisions 6-8 below.

[3] See Appendix 1 Map 7 Okanagan Timber Supply Area

[4] See Appendix 1 Map 8 Merritt Timber Supply Area

5. **Potential reduction in old forest retention in low biodiversity emphasis areas**

For landscape units with a low biodiversity emphasis, the old forest retention percent listed in Tables 1-4 may be reduced by up to 2/3, to the extent necessary to address impacts on timber supply.[5]

6. **Use of younger forests to meet old forest objectives**

In intermediate and high emphasis landscape units where it can be demonstrated that equal or better conservation benefits would result, stands less than the age of old, and preferably mature forest, may contribute to the percentage of old forest retention defined in Tables 1-4.[6]

In the Okanagan and Merritt areas where it can be demonstrated that equal or better conservation benefits would result, younger stands and preferably mature forest, can be used to meet the area requirements listed in Tables 1 and 2, Appendix 2.

7. **Recruitment and catastrophic natural disturbances**

In landscape units where there is:

a. An immediate recruitment situation due to insufficient old forest in a variant; or

b. An epidemic or catastrophic natural event such as wildfire or insects

A recruitment strategy approved by the Minister of Sustainable Resource Management or his delegate[7] can be used to meet the requirements of Objective 2, 3 and 4.[8]

8. **Draft old growth management areas**

Where the Minister of Sustainable Resource Management, his delegate, or a licensee or group of licensees has identified draft old growth management areas, the Minister or delegate may specify in writing that these draft old growth management areas meet the intent of this Order[9].

9. **Pilot project regulation areas**

For the area encompassed by the Fort St John, Riverside and Stillwater pilot project regulations, the Minister of Sustainable Resource Management or his delegate may specify in writing the sections of the approved Sustainable Forest Management Plan[10], or other approved operational plan as required under the pilot regulation which meets the intent of this Order.

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[5] See Implementation Policy for further guidance on reductions in percent retention in low biodiversity emphasis areas.
[6] See Implementation Policy for further technical information regarding the use of younger stands.
[7] Delegate is defined in the delegation memo from the Minister of SRM to regional directors July 2002 Reference: 29048, and as will be amended from time to time.
[8] See Implementation Policy for guidance on recruitment.
[9] This requires that the draft old growth management areas are retained or managed for old forest values.
[10] Approved is defined in section 39 of the Fort St John Pilot Project regulation.

B. <u>Effective date and future orders</u>

This Order comes into effect June 30, 2004.

For the purposes of the *Forest Practices Code of British Columbia Act*, all new forest development plans and all major amendments to forest development plans submitted 4 months after the effective date of this Order must be consistent with the Order.

This Order does not affect Category A cutblocks approved pursuant to the *Forest Practices Code of British Columbia Act* on or before the effective date.

For the purposes of the *Forest and Range Practices Act*, and despite subsection 16(2) of that Act, all forest stewardship plans submitted after the effective date must be consistent with this Order.

When a new order of the Minister of Sustainable Resource Management or delegate establishes old forest objectives, this Order will, on the effective date of the new order, cease to have effect for the area or areas affected by old forest established in the new order.

Where a new order of the Minister of Sustainable Resource Management or delegate establishes old forest objectives for a portion of a landscape unit that is established under this Order, this Order remains in effect for the portion of the landscape unit for which the new order does not establish old forest objectives.

C. <u>Continuous effect of previously established old growth objectives</u>

This Order does not apply to spatially located old growth management areas or non-spatial old growth objectives established in a higher-level plan prior to the effective date of this Order under either section 3 or section 4 of the *Forest Practices Code of British Columbia Act.*

D. <u>Review of the Order</u>

The Minister of Sustainable Resource Management recognizes that new information will emerge regarding the role of old forest in the maintenance of biodiversity, the efficacy of the objectives in this Order in that regard, and the economic and social implications of old forest and biodiversity conservation. The minister therefore will review this Order no later than March 31, 2007 to assess its effectiveness in achieving government's sustainable resource management goals.

The Honourable George Abbott
Minister of Sustainable Resource Management

APPENDIX 1
TABLE 1

LANDSCAPE UNIT TABLE

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
2	Aaltanhash	Low
3	Adam-Eve	Low
6	Ahnuhati-kwalate	High
7	Ahta	High
8	Aiken	Intermediate
9	Ainslie	High
1469	Akie	Low
1471	Akie River	Low
11	Akolkolex	Multiple
1472	Akue	Intermediate
20	Allison	Low
24	Alouette	Intermediate
29	Anarchist	Multiple
30	Anderson	Intermediate
33	Anstey	Multiple
35	Anyox	Low
1473	Anzac	High
37	Ape	Low
38	Aristazabal	Intermediate
39	Artlish	Intermediate
40	Ash	Intermediate
42	Ashnola	High
1474	Asitka	Intermediate
44	Athlow Bay	Low
45	Atlin Lake	High
46	Atna	Low
1475	Averil	Low
1450	Babine	Intermediate
51	Babine	Intermediate
52	Babine East	Intermediate
54	Babine River	High
55	Babine West	Low
60	Banks	Intermediate
61	Barkley Sound Islands	Low
230	Barrington River	Intermediate
1476	Bastille	Intermediate
69	Bear	Intermediate
1477	Bearhole	Intermediate
71	Beaver	Intermediate
1478	Belcourt	Intermediate
79	Belize	Intermediate
80	Bella Coola	Intermediate
81	Belle Bay	Intermediate
82	Beresford	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1479	Big Beaver	Intermediate
92	Big Eddy	Multiple
93	Big Falls	Low
95	Big Silver	Intermediate
100	Bigmouth	Low
1481	Bill's	Low
103	Billygoat	Intermediate
1482	Birdflat	Low
104	Birkenhead	High
106	Bishop	Intermediate
105	Bishop	Low
1485	Blackwater	Intermediate
1483	Blackwater	Low
117	Blue River	High
1486	Blueberry	Low
1487	Bluff Creek	High
122	Blunt	Low
125	Bonanza	Intermediate
1488	Boreal	Intermediate
1489	Boucher	Low
1490	Bowron	Intermediate
132	Bowser	Low
133	Braden	Low
1491	Braid	High
136	Brem	Intermediate
140	Bridge	Low
143	Brittain	Intermediate
145	Brooks	Low
146	Broughton	Low
147	Brown	Low
148	Brown Bear	Low
149	Buck	Low
1492	Buffalohead	Low
154	Bulkley	Intermediate
155	Bulkley	Intermediate
1493	Bunch	Low
157	Bunster	Intermediate
158	Burman	Low
156	Burnie	High
159	Burnie	High
160	Burns Lake East	Low
161	Burns Lake West	Low
1494	Burnt - Lemoray	Intermediate
167	Bute East	Intermediate
168	Bute West	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
169	Butedale	Intermediate
170	Buttle	Low
172	Callaghan	Low
173	Calvert	Low
174	Cambria Icefield	Low
175	Cameron	Intermediate
176	Campania	Intermediate
1495	Canoe	Low
1496	Capot Blanc	Intermediate
1497	Captain	Intermediate
183	Captain	Low
1498	Carbon	Intermediate
1499	Cariboo	Low
1500	Carp	Intermediate
191	Carpenter Lake North	Low
192	Carpenter Lake South	Multiple
1501	Castle	Low
1502	Catkin	Intermediate
196	Caycuse	Intermediate
199	Chambers	Intermediate
200	Chapman	Low
201	Chapman	Low
202	Chapple	Low
203	Charles	Intermediate
1504	Chee	Intermediate
207	Chehalis	Intermediate
208	Chelaslie	High
209	Chemainus	Intermediate
1817	Cherryville	Low
212	Cheslatta	Intermediate
1506	Chilako	Intermediate
1505	Chilako	Low
218	Chilliwack	Low
220	China	Intermediate
1507	Chuchi	Intermediate
227	Chukachida River	Intermediate
1508	Chunamon	Low
232	Clayton	Low
1509	Clearwater	Intermediate
237	Clore	Intermediate
1510	Cluculz	Low
243	Clyak	High
1511	Coal	Low
252	Coldwater	Multiple
1512	Collins - Davis	Low
1513	Connaghan Creek	High
256	Connel Creek	Multiple
258	Copper	Intermediate
259	Coquihalla	Intermediate
260	Coquitlam	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
263	Corrigan	Intermediate
264	Cortes	Intermediate
265	Corya	High
266	Cous	Low
267	Cowichan	Low
268	Crab	Low
269	Crag	Low
273	Cranberry	Multiple
270	Cranberry River	Intermediate
276	Craven	Intermediate
1514	Crehan	Intermediate
1515	Crescent Spur	High
1516	Crooked	Intermediate
1517	Crow	Intermediate
283	Crowfoot	Low
1518	Crusty	Intermediate
285	Cry Lake	Intermediate
1519	Crying Girl	Intermediate
1520	Cunningham	Low
1521	D Easum	Low
291	Dala	Low
295	Dasque	Low
1524	Dawson	Low
1525	Dawson Creek	Low
302	Dean	Intermediate
303	Dease Lake	Intermediate
304	Dease River	High
307	Deep Canoe	Low
308	Deep Creek	Low
310	Denny	Low
1816	Deserted	High
1526	Dilly	Low
1527	Discovery	Intermediate
1529	Dome	High
321	Don Peninsula	Intermediate
322	Doos/Dallery	Low
1530	Dore	Low
327	Downie	Multiple
330	Draney	Low
1531	Driftwood	Intermediate
1532	Duckling	Intermediate
333	Duffey Lake	High
336	Dundas	Low
1533	Dunedin	High
1534	Dunkley	Low
1535	Dunlevey	High
1536	Dust	Low
1537	Duti	High
345	Eagle River	Intermediate
355	East Harrison	Low
356	East Howe	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1538	East Kinbasket	Low
1539	East Pine	Low
1542	EastTwin-McKale	Low
361	Eden Lake	Intermediate
363	Effingham	Intermediate
1544	Eight Mile	Low
1545	Eklund	High
365	Elaho	Intermediate
367	Eliza	Low
1546	Elleh	Low
369	Ellerslie	Low
1548	Endako	Low
374	Englishman	Intermediate
1549	Entiako	High
378	Escalante	Low
1550	Eskai	Low
381	Estero	Low
1551	Etane	Intermediate
385	Evans	Low
386	Exchamsiks	Low
387	Exstew	Intermediate
1552	Falk	Intermediate
1553	Fall	High
390	Falls	Low
1555	Firth	Intermediate
401	Fish Egg	Low
403	Fleming	Intermediate
405	Foch	High
1556	Fontinako	High
1557	Forcier	Low
1558	Forgetmenot	Intermediate
1559	Fort Nelson River A	Low
1560	Fort Nelson River B	Low
1561	Foster	Low
1562	Fox	High
1563	Framstead	Intermediate
424	Francois East	Low
425	Francois West	Intermediate
426	Franklin	High
1564	Fraser	Intermediate
429	Fraser Valley South	Low
1827	French	Multiple
430	French Bar	Intermediate
431	French Creek	Low
434	Frisby Ridge	Multiple
1565	Frog	High
1567	Frypan	Low
438	Fulmore	Intermediate
439	Fulton	Low
1568	Gaffney	Low
442	Gail	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1569	Gammer	Intermediate
450	Gates	Intermediate
1570	Gathto	Low
1571	Gemini	Intermediate
1572	Germansen Mountain	Low
1573	Gething	Low
456	Gil	Low
457	Gilford	Low
1574	Gillis	Intermediate
1467	Gilttoyees	Intermediate
468	Gladys River	High
1577	Gleason	Intermediate
1578	Goat	Intermediate
473	Gold	High
477	Goldstream	Multiple
479	Gordon	Intermediate
480	Gosnel	Intermediate
1579	Graham	High
485	Granisle	Low
489	Gray	Low
1580	Graybank	Low
1581	Grayling	Low
495	Great Central	Intermediate
496	Green	Intermediate
499	Greenville	Low
498	Greenville - Kalum	Intermediate
1582	Gregg	Low
502	Gribbell	Low
1583	Grizzly	Intermediate
1584	Grostete	Intermediate
505	Gudal	Low
509	Gun	Low
1585	Gundahoo	Intermediate
1586	Gwillim	Intermediate
1587	Haggen	Intermediate
1588	Halett	Intermediate
1589	Halfway	High
520	Hanawald	Low
521	Harold Price	Intermediate
522	Harris	Intermediate
523	Hartley	Intermediate
524	Haslam	Low
527	Hatzic	Low
528	Hawkes	Intermediate
529	Hawkesbury Island East	Low
530	Hawkesbury Island West	Intermediate
1590	Hay River	Intermediate
536	Hayes	Multiple
537	Hazelton	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
543	Helmcken	Intermediate
544	Henderson	Low
547	Hevenor	High
1592	Hewer	Intermediate
549	Hibben	High
1593	Highhat	Low
552	Hirsch	Low
1594	Hixon	Intermediate
1595	Hoffard	Intermediate
556	Holberg	Low
1596	Holden	Low
1597	Holmes	Intermediate
560	Homathko	High
561	Homfray	Intermediate
562	Honna	Intermediate
564	Horetzky	Low
568	Horne	Multiple
1598	Horsey-Small	Low
1599	Hossitl	Low
576	Hot Springs	Low
578	Houston - Tommy	High
579	Howe	Intermediate
583	Huaskin	Intermediate
1600	Hudson's Hope	Low
1601	Hugh Allan	Intermediate
1602	Humbug	High
586	Hunter	Low
587	Hurley	Intermediate
588	Ian	Intermediate
590	Iknouk	Intermediate
593	Illecillewaet	Multiple
1603	Imperial - Monkman	High
595	Indian	Low
1604	Ingenika	Intermediate
597	Inklin River	High
598	Intata	Intermediate
1605	Inzana	Low
1606	Irene E	Intermediate
1607	Irene W	High
604	Ishkheenickh	Intermediate
1608	Jackfish	High
1613	Jackknife	Low
608	Jalun	Low
1614	Jarvis	High
611	Jennings River	Intermediate
1615	Jerryboy	Low
615	Jervis	Intermediate
616	Jesse - Bish	Low
1616	Johanson	Intermediate
620	Johnston	Intermediate
619	Johnston	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
624	Jordan	Multiple
1617	July Lake	Low
627	Jump Across	Low
628	Juniper	Low
1618	Kahntah	Intermediate
629	Kaien	Low
630	Kakiddi Creek	Intermediate
631	Kakweiken	Intermediate
632	Kalum	Intermediate
633	Kaouk	Intermediate
634	Kashutl	Low
1466	Kasiks	Intermediate
1619	Kechika River	High
642	Kemano	Intermediate
1620	Kennedy	High
1621	Kenneth	Intermediate
242	Keogh	Low
645	Keremeos	Intermediate
647	Kettle	Low
648	Khtada	High
649	Khutze	Intermediate
1464	Khutzeymateen Park	Intermediate
650	Khyex	Low
829	Kidprice	Intermediate
655	Kilbella/Chuckwalla	Intermediate
656	Kilippi	Low
657	Kiltuish	Low
663	King Island	Low
664	Kingfisher	Intermediate
665	Kinskuch	Intermediate
1623	Kinuseo	Intermediate
1624	Kiskatinaw	Low
668	Kispiox	Low
1625	Kitchi	High
670	Kiteen	Low
671	Kitimat	Low
672	Kitkiata	High
673	Kitsault	Intermediate
675	Kitseguecla	High
674	Kitseguecla	Low
676	Kitwancool	Low
677	Kitwanga	Low
1626	Kitza	Intermediate
1627	Kiwa-Tete	Low
1628	Kiwigana	Intermediate
681	Klanawa	Intermediate
682	Klappan River	Intermediate
683	Klaskish	High
1629	Klawli	Intermediate
1630	Klawli	Intermediate
686	Kleanza - Treasure	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1631	Kledo	Intermediate
688	Kleeptee	Low
689	Klekane	Low
691	Klinaklini Glacier	Intermediate
1632	Klowee	Intermediate
1633	Klua	Intermediate
1634	Kluatantan	Intermediate
1635	Kluskus	Low
697	Knight East	Low
1636	Kobes	Intermediate
700	Koksilah	Low
706	Kowesas	Low
707	Ksedin	Low
708	Kshwan	High
709	Kuldo	Low
710	Kumealon	High
712	Kusawa River	Intermediate
717	Kwatna/Quatlena	Intermediate
719	Kwinamass	Intermediate
720	Kwinamuck	Low
718	Kwoiek	Multiple
1639	Kwokullie	Intermediate
1640	Kyklo	Low
1641	La Biche	Intermediate
725	Labouchere	Low
730	LaForme	Multiple
732	Lakelse	Intermediate
735	Laredo	Intermediate
736	Larkworthy	Low
1648	Liard Hot Springs	Intermediate
1649	Liard River A	Intermediate
1650	Liard River B	Intermediate
1651	Liard River C	High
746	Liberty	Multiple
1652	Lion	Low
752	Little Qualicum	Intermediate
753	Little Rancheria River	Intermediate
755	Lizzie	Intermediate
759	Lois	Low
762	Loss	Low
763	Lost Creek	High
768	Louise Island	Low
1653	Lovell	Low
1654	Lower Akie	High
1655	Lower Beatton	Low
780	Lower Iskut River	Intermediate
781	Lower Kechika River	Intermediate
782	Lower Kimsquit	High
783	Lower Kingcome	Intermediate
784	Lower Klinaklini	Intermediate
1656	Lower Moberly	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1657	Lower Morkill/Cushing	Intermediate
788	Lower Nicola	Multiple
789	Lower Nimpkish	Low
1658	Lower Ospika	Intermediate
1659	Lower Pesika	High
796	Lower Squamish	Intermediate
797	Lower Stikine River	Intermediate
800	Lower Yakoun	Intermediate
1660	Lucas	Intermediate
802	Lull-Sallie	Intermediate
809	Mabel	Low
1661	MacDonald	Intermediate
811	Machmell	Intermediate
814	Madely	Intermediate
815	Maggie	Low
816	Mahatta	Low
1662	Major Hart	Intermediate
818	Malcolm	Low
819	Mamquam	Low
820	Manning	Intermediate
1663	Manson River	Low
822	Marble	Intermediate
823	Marmot	Low
1664	Martin Creek	Low
825	Masset Inlet	Low
1665	Matulka	Intermediate
1666	McBride-Dunster	Low
831	McCauley	Low
833	McCully	Intermediate
1667	McCusker	High
1668	Mcleod	Low
842	McNulty	Multiple
844	Meager	Intermediate
847	Mehatl	Intermediate
1669	Mesilinka	Low
850	Mess Creek	Intermediate
855	Mica	Multiple
1670	Middle	Intermediate
857	Middle Dease River	High
858	Middle Iskut River	Intermediate
859	Middle Klinaklini	Intermediate
862	Middle Stikine River	Intermediate
1671	Milk	Low
1672	Milligan	Intermediate
869	Millstone	Low
1673	Minaker	Intermediate
872	Miriam	Intermediate
1674	Misinchinka	Low
1675	Missinka	Intermediate
875	Mission	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1676	Mollie	Low
880	Monckton	Low
882	Moonlit	Low
1677	Moose	Intermediate
1678	Morfee	Intermediate
886	Morice Lake	High
887	Morrison	Intermediate
1681	Mosque	Low
1682	Moule	Low
1684	Mud	Intermediate
897	Mulvehill	Multiple
1685	Muncho	Low
900	Murray	Low
901	Muskaboo	Low
1687	Muskeg	Low
1688	Muskwa River A	Low
1689	Muskwa River B	Low
1690	Nabesche	Intermediate
907	Nadina	Intermediate
908	Nahatlatch	High
909	Nahline River	Intermediate
910	Nahmint	High
911	Nahwitti	Intermediate
912	Naikoon	Low
913	Naka	Low
914	Nakina River	High
915	Nanaimo	Intermediate
916	Nanika	High
917	Nanoose	Low
1691	Narraway	High
920	Narrows	Intermediate
921	Nascall	Low
922	Nasparti	Low
923	Nass	Intermediate
924	Nass River Kalum	High
1693	Nation	High
1692	Nation	Low
925	Natlan	Low
1694	Nechako	Low
1695	Nechako	Low
932	Neechanz	High
934	Nekite	High
935	Nelson - Fiddler	Low
937	Neroutsos	Low
1696	Netson	Intermediate
943	Nigei	Low
944	Nilkitkwa	High
1697	Nina Creek	High
1699	Nithi	Low
953	Nitinat	Intermediate
954	Nootka	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
955	Nootum/Koeye	Intermediate
956	North Babine	Low
1700	North Firesteel	High
1701	North Ingenika	High
1702	Northern Trench	Intermediate
964	Nusatsum	Intermediate
1703	Obo River	High
966	Observatory East	Low
967	Observatory West	Low
970	Okanagan West Side	Multiple
971	Olh	Low
1704	Omineca	High
1706	Ominicetla	High
1707	One Island	Low
975	Ootsa	Intermediate
1708	Ootta	Intermediate
1710	Osilinka	Low
1712	Otelsas	Intermediate
982	Otter	Multiple
983	Otun	Low
984	Outer Coast Islands	Low
1713	Ovington	Intermediate
986	Oweegee	Low
987	Owen	Intermediate
988	Owikeno	Low
1044	Oyster	Low
989	Pa - aat	Intermediate
996	Parrott	Low
1716	Parsnip	High
1714	Parsnip	Intermediate
1717	Patry	Intermediate
1000	Pavilion	Intermediate
1002	Pearse	Low
1718	Pelly	High
1013	Pennask	Intermediate
1016	Penticton	Intermediate
1719	Pesika	Intermediate
1720	Petitot River	Intermediate
1724	Philip	Low
1725	Philip Lake	Intermediate
1024	Phillips	High
1726	Pinchi	Intermediate
1727	Pine Pass	Intermediate
1729	Pine River	Low
1028	Pingston	Multiple
1029	Pitman River	Intermediate
1030	Pitt	Intermediate
1032	Porcher	Low
1730	Pouce	Low
1034	Powell Daniels	Intermediate
1035	Powell Lake	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1039	Price	Low
1731	Prince	Intermediate
1732	Prophet River	Low
1733	Puggins	Low
1043	Pukeashun	Intermediate
1734	Punchaw	Intermediate
1046	Puntledge	Low
1735	Purden	Low
1050	Quadra	Intermediate
1054	Quatam	Intermediate
1058	Quinsam	Low
1059	Quottoon	Intermediate
1736	Rabbit	Low
1737	Racing	Low
1063	Railroad	High
1738	Ram	Low
1739	Raush	Intermediate
1068	Red Bluff	Low
1740	Redpott	Intermediate
1829	Redrock	Multiple
1741	Redwillow	High
1073	Reiseter	Intermediate
1077	Rennell	Intermediate
1742	Reynolds	High
1743	Richards	Low
1086	Roderick	Low
1087	Rogers	Intermediate
1088	Roscoe	Low
1089	Rosewall	Intermediate
1092	Ryan	Low
1744	Sahtaneh	Low
1094	Sallysout	Low
1745	Salmon	Low
1095	Salmon	Low
1098	Salmon Arm	Intermediate
1099	Salmon Inlet	Intermediate
1100	Saloompt	Intermediate
1101	San Josef	Intermediate
1102	San Juan	Intermediate
1746	Sandy	Intermediate
1106	Sarita	Low
1107	Sayward	Intermediate
1747	Scatter	Intermediate
1748	Schooler	Intermediate
1110	Scotia	Low
1112	Sechelt	Low
1749	Seebach	Low
1750	Selwyn	High
1754	Septimus	Intermediate
1116	Seven Sisters	Low
1117	Sewell	Intermediate

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1825	Seymour	High
1118	Seymour	Intermediate
1120	Seymour-Capilano	Intermediate
1755	Sharktooth	Low
1122	Shawnigan	Low
1123	Shedin	Low
1124	Sheemahant	Intermediate
1125	Sheep Passage	Low
1129	Shegunia	Intermediate
1756	Shekilie	Low
1131	Sheladamus	Intermediate
1132	Shelagyote	High
1133	Sheslay River	Intermediate
1134	Shushartie	High
1136	Sibola	High
1137	Sicintine	Intermediate
1757	Sikanni	High
1140	Silverhope	Intermediate
1141	Sim	Intermediate
1144	Similkameen	Low
1830	Similkameen	Multiple
1145	Simms	Low
1146	Siska	Intermediate
1758	Skeena	Intermediate
1151	Skeena Crossing	Intermediate
1150	Skeena Islands	High
1153	Skeena River Kalum	High
1154	Skeena West	Low
1156	Skidegate Lake	Low
1162	Skwawka	High
1759	Slamgeesh	Intermediate
1760	Slender	Low
1761	Slim	High
1168	Sloquet	Intermediate
1170	Sloquet	Intermediate
1167	Sloquet-High	High
1762	Smith	Intermediate
1172	Smith - Willis	Multiple
1169	Smith Sound	Low
1173	Smitley/Noeick	Intermediate
1174	Smokehouse	Low
1763	Snake	Intermediate
1177	Snowdrift	Intermediate
1181	Soards	Multiple
1182	Somass	Low
1183	Somerville	Intermediate
1184	Soo	Low
1185	Sooke	Low
1186	South Bentinck	Low
1764	South Firesteel	High
1765	South Germansen -	High

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
	Upper Manson	
1190	Southgate	High
1767	SouthTrench	Intermediate
1768	Spakwaniko	High
1194	Sparkling	Intermediate
1769	Spatsizi	Intermediate
1195	Spatsizi River	Intermediate
1198	Spius	Multiple
1199	Sproat Lake	Intermediate
1821	Spruce Lake	Intermediate
1201	Spuzzum	Low
1770	Squingula	Low
1204	Stafford	High
1206	Stagoo	Intermediate
1771	Stanolind	Low
1208	Stave	Low
1213	Stephens	Low
1214	Stikine River	Intermediate
1772	Stony	Intermediate
1775	Stuart	Intermediate
1773	Stuart	Low
1774	Stuart	Low
1776	Sulpher	Low
23	Summers	Multiple
1226	Sumquolt	Intermediate
1227	Surf	Intermediate
1228	Suskwa	Intermediate
1777	Sustut	Intermediate
1778	Sutherland	Intermediate
1233	Sutslem/Skowquiltz	Low
1822	Swakum	Multiple
1779	Swannell	High
1237	Sweetin	Intermediate
1239	Swift River	Intermediate
1240	Swindle	Low
1780	Table	Intermediate
1781	Tachick	Intermediate
1244	Tahltan River	Intermediate
1245	Tahsis	Low
1246	Tahsish	Intermediate
1247	Tahtsa	Low
1782	Takla	Intermediate
1248	Talchako/Gyllenspetz	Intermediate
1249	Taleomey/Asseek	Intermediate
1250	Taltapin	Low
1252	Tasu	Low
1783	Tatelkuz	Intermediate
1256	Tatshenshini River	Intermediate
1258	Taylor - Damdochax	Intermediate
1785	Tchentlo	Intermediate
1261	Tchitin	High

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1264	Telkwa	Intermediate
1266	Tenas	Intermediate
1786	Tentsi	Intermediate
1268	Teslin River	Intermediate
1787	Tetsa	Low
1271	Texada	Low
1272	Texas Creek	Intermediate
1788	Tezzeron	Low
1789	TFL42	Low
1275	Thautil	High
1277	Thurlow	Low
1790	Thutade	High
1791	Timberwolf	Intermediate
1282	Tintina	Low
1283	Tlell	Intermediate
1284	Tlupana	Intermediate
1285	Toba	High
1288	Tochcha - Natowite	Intermediate
1291	Tolmie	High
1792	Tommy Lakes	Low
1292	Topley	Low
1293	Toquart	Intermediate
1294	Torkelson	Low
1793	Torpy	Intermediate
1298	Trent	Low
1299	Trepanier	Multiple
1302	Tretheway	Intermediate
1304	Trinity	Low
1305	Triotsa	High
1306	Triumph	Low
1309	Trout	Intermediate
1307	Trout Creek	Intermediate
1794	Trutch	Intermediate
1310	Trutch	Low
1313	Tseaux	Intermediate
1314	Tsitika	High
1048	Tsolum	Low
1315	Tsulquate	Intermediate
1795	Tuchodi	Low
1317	Tuck	Low
1796	Tudyah	Intermediate
1319	Tugwell	Low
1320	Tulameen	Multiple
1322	Turnagain River	Intermediate
1798	Tutizza	High
1324	Tutshi River	Intermediate
1325	Tuwasus	Intermediate
1326	Tuya River	Intermediate
1799	Twenty Mile	Intermediate
1330	Twin	Low
1333	Union	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1334	Unuk River	Intermediate
1800	Upper Akie River	High
1339	Upper Campbell	Low
1343	Upper Cranberry	Low
1801	Upper Gataga	High
1352	Upper Iskut River	Intermediate
1353	Upper Jennings River	Intermediate
1355	Upper Kettle	Low
1356	Upper Kimsquit	Intermediate
1357	Upper Kingcome	Intermediate
1358	Upper Kispiox	Low
1359	Upper Klinaklini	Intermediate
1360	Upper Liard River	Intermediate
1361	Upper Lillooet	Intermediate
1802	Upper Moberly	Intermediate
1803	Upper Morkill	Intermediate
1823	Upper Nicola	Multiple
1367	Upper Nimpkish	Intermediate
1804	Upper Ospika	High
1805	Upper Pelly	High
1372	Upper Salmon	Low
1826	Upper Shuswap	High
1378	Upper Squamish	Low
1380	Upper Stikine River	Intermediate
1806	Upper Sukunka	Low
1385	Valley	Intermediate
1807	Vents	Low
1389	Vernon	Low

Landscape Unit (LU) Identifier	Landscape Unit (LU) Name	Biodiversity Emphasis (BE)
1393	Wakeman	Intermediate
1394	Walbran	Intermediate
1808	Wapiti	Intermediate
1400	Washwash	Low
1402	Watson Bar	Low
1403	Wedeene	Intermediate
1412	West Harrison	Low
1809	West Kinbasket	Low
1418	Whalen	Intermediate
1419	Whistler	Low
1421	White	High
1422	White	Intermediate
1423	White	Low
1810	Whitefish	Intermediate
1425	Whitesail	Intermediate
1811	Wicked River	High
1428	Widgeon	Intermediate
1430	Wildfire	Intermediate
1813	Willow	Low
1814	Wolverine	Intermediate
1815	Woodall	Low
1444	Yakoun Lake	High
1445	Yalakom	Intermediate
1446	Yale	Low
1447	Yeo	Low
1449	Zeballos	Low

Important Note: The following maps are displayed for information only, and should not be used to locate landscape unit boundaries. Landscape unit boundaries are available on maps at regional offices of the Ministry of Sustainable Resource Management.[11]

Map 1	**Overview**
Map 2	**Omineca-Peace Area**
Map 3	**Skeena Area**
Map 4	**Revelstoke Timber Supply Area**
Map 5	**Lillooet Timber Supply Area**
Map 6	**Coast Area**
Map 7	**Okanagan Timber Supply Area**
Map 8	**Merritt Timber Supply Area**

[11] Landscape unit boundaries are available on maps at regional offices of Sustainable Resource Management, as may be amended from time to time by the regional director of Sustainable Resource Management.

Appendix 3

A Table of Biogeoclimatic Units assigned to Natural Disturbance Types

Contains all units mapped in QBEC_BC version 5 (April 2003)[15]

NDT	BECLABEL	Zone	Subzone	Variant	Phase	SubzoneName	VariantName	PhaseName
5	**AT un**	AT	un			Undefined		
5	**AT unp**	AT	unp			Undefined Parkland		
4	**BG xh 1**	BG	xh	1		Very Dry Hot	Okanagan	
4	**BG xh 2**	BG	xh	2		Very Dry Hot	Thompson	
4	**BG xh 3**	BG	xh	3		Very Dry Hot	Fraser	
4	**BG xw 1**	BG	xw	1		Very Dry Warm	Nicola	
4	**BG xw 2**	BG	xw	2		Very Dry Warm	Alkali	
3	**BWBSdk 1**	BWBS	dk	1		Dry Cool	Stikine	
3	**BWBSdk 2**	BWBS	dk	2		Dry Cool	Liard	
3	**BWBSmw 1**	BWBS	mw	1		Moist Warm	Peace	
3	**BWBSmw 2**	BWBS	mw	2		Moist Warm	Fort Nelson	
3	**BWBSun**	BWBS	un			Undefined		
3	**BWBSvk**	BWBS	vk			Very Wet Cool		
3	**BWBSwk 1**	BWBS	wk	1		Wet Cool	Murray	
3	**BWBSwk 2**	BWBS	wk	2		Wet Cool	Graham	
3	**BWBSwk 3**	BWBS	wk	3		Wet Cool	Kledo	
2	**CDF mm**	CDF	mm			Moist Maritime		
2	**CWH dm**	CWH	dm			Dry Maritime		
2	**CWH ds 1**	CWH	ds	1		Dry Submaritime	Southern	
2	**CWH ds 2**	CWH	ds	2		Dry Submaritime	Central	
2	**CWH mm 1**	CWH	mm	1		Moist Maritime	Submontane	
2	**CWH mm 2**	CWH	mm	2		Moist Maritime	Montane	
2	**CWH ms 1**	CWH	ms	1		Moist Submaritime	Southern	
2	**CWH ms 2**	CWH	ms	2		Moist Submaritime	Central	
1	**CWH un**	CWH	un			Undefined		
1	**CWH vh 1**	CWH	vh	1		Very Wet Hypermaritime	Southern	
1	**CWH vh 2**	CWH	vh	2		Very Wet Hypermaritime	Central	
1	**CWH vm**	CWH	vm			Very Wet Hypermaritime	Undifferentiated	
1	**CWH vm 1**	CWH	vm	1		Very Wet Maritime	Submontane	
1	**CWH vm 2**	CWH	vm	2		Very Wet Maritime	Montane	
1	**CWH vm 3**	CWH	vm	3		Very Wet Maritime	Central	
1	**CWH wh 1**	CWH	wh	1		Wet Hypermaritime	Submontane	

[15] This list and NDT map is posted at: http://www.for.gov.bc.ca/ftp/hre/external/!publish/becmaps

1	**CWH wh 2**	CWH	wh	2	Wet Hypermaritime	Montane
1	**CWH wm**	CWH	wm		Wet Maritime	
2	**CWH ws 1**	CWH	ws	1	Wet Submaritime	Submontane
2	**CWH ws 2**	CWH	ws	2	Wet Submaritime	Montane
2	**CWH xm 1**	CWH	xm	1	Very Dry Maritime	Eastern
2	**CWH xm 2**	CWH	xm	2	Very Dry Maritime	Western
3	**ESSFdc 1**	ESSF	dc	1	Dry Cold	Okanagan
3	**ESSFdc 2**	ESSF	dc	2	Dry Cold	Thompson
5	**ESSFdcp**	ESSF	dcp		Dry Cold Parkland	
3	**ESSFdcw**	ESSF	dcw		Dry Cold Woodland	
3	**ESSFdk**	ESSF	dk		Dry Cool	
5	**ESSFdkp**	ESSF	dkp		Dry Cool Parkland	
3	**ESSFdkw**	ESSF	dkw		Dry Cool Woodland	
3	**ESSFdm 1**	ESSF	dm	1	Dry Mild	Southeast Kootenay
5	**ESSFdmp**	ESSF	dmp		Dry Mild Parkland	
3	**ESSFdmw**	ESSF	dmw		Dry Mild Woodland	
3	**ESSFdv**	ESSF	dv		Dry Very Cold	
5	**ESSFdvp**	ESSF	dvp		Dry Very Cold Parkland	
2	**ESSFmc**	ESSF	mc		Moist Cold	
5	**ESSFmcp**	ESSF	mcp		Moist Cold Parkland	
2	**ESSFmk**	ESSF	mk		Moist Cool	
5	**ESSFmkp**	ESSF	mkp		Moist Cool Parkland	
2	**ESSFmm 1**	ESSF	mm	1	Moist Mild	Raush
2	**ESSFmm 2**	ESSF	mm	2	Moist Mild	Robson
5	**ESSFmmp**	ESSF	mmp		Moist Mild Parkland	
2	**ESSFmv 1**	ESSF	mv	1	Moist Very Cold	Nechako
2	**ESSFmv 2**	ESSF	mv	2	Moist Very Cold	Bullmoose
2	**ESSFmv 3**	ESSF	mv	3	Moist Very Cold	Omineca
2	**ESSFmv 4**	ESSF	mv	4	Moist Very Cold	Graham
5	**ESSFmvp**	ESSF	mvp		Moist Very Cold Parkland	
2	**ESSFmw**	ESSF	mw		Moist Warm	
5	**ESSFmwp**	ESSF	mwp		Moist Warm Parkland	
1	**ESSFvc**	ESSF	vc		Very Wet Cold	
5	**ESSFvcp**	ESSF	vcp		Very Wet Cold Parkland	
1	**ESSFvv**	ESSF	vv		Very Wet Very Cold	
5	**ESSFvvp**	ESSF	vvp		Very Wet Very Cold Parkland	
1	**ESSFwc 1**	ESSF	wc	1	Wet Cold	Columbia
1	**ESSFwc 2**	ESSF	wc	2	Wet Cold	Northern
1	**ESSFwc 3**	ESSF	wc	3	Wet Cold	Cariboo
1	**ESSFwc 4**	ESSF	wc	4	Wet Cold	Selkirk
2	**ESSFwc 5**	ESSF	wc	5	Wet Cold	Salmo

2	**ESSFwc 6**	ESSF	wc	6		Wet Cold	Ymir	
5	**ESSFwcp**	ESSF	wcp			Wet Cold Parkland		
2	**ESSFwcw**	ESSF	wcw			Wet Cold Woodland		
1	**ESSFwk 1**	ESSF	wk	1		Wet Cool	Cariboo	
1	**ESSFwk 2**	ESSF	wk	2		Wet Cool	Misinchinka	
5	**ESSFwkp**	ESSF	wkp			Wet Cool Parkland		
1	**ESSFwm**	ESSF	wm			Wet Mild		
5	**ESSFwmp**	ESSF	wmp			Wet Mild Parkland		
1	**ESSFwmw**	ESSF	wmw			Wet Mild Woodland		
1	**ESSFwv**	ESSF	wv			Wet Very Cold		
5	**ESSFwvp**	ESSF	wvp			Wet Very Cold Parkland		
3	**ESSFxc**	ESSF	xc			Very Dry Cold		
5	**ESSFxcp**	ESSF	xcp			Very Dry Cold Parkland		
2	**ESSFxv 1**	ESSF	xv	1		Very Dry Very Cold		
2	**ESSFxv 2**	ESSF	xv	2		Very Dry Very Cold		
3	**ICH dk**	ICH	dk			Dry Cool		
3	**ICH dm**	ICH	dm			Dry Mild		
3	**ICH dw 1**	ICH	dw	1		Dry Warm	West Kootenay	
3	**ICH dw 2**	ICH	dw	2		Dry Warm	Boundary	
2	**ICH mc 1**	ICH	mc	1		Moist Cold	Nass	
2	**ICH mc 2**	ICH	mc	2		Moist Cold	Hazelton	
3	**ICH mk 1**	ICH	mk	1		Moist Cool	Kootenay	
3	**ICH mk 2**	ICH	mk	2		Moist Cool	Thompson	
2	**ICH mk 3**	ICH	mk	3		Moist Cool	Horsefly	
2	**ICH mm**	ICH	mm			Moist Mild		
2	**ICH mw 1**	ICH	mw	1		Moist Warm	Golden	
2	**ICH mw 2**	ICH	mw	2		Moist Warm	Columbia-Shuswap	
3	**ICH mw 3**	ICH	mw	3		Moist Warm	Thompson	
2	**ICH mw 4**	ICH	mw	4		Moist Warm	Ymir	
1	**ICH vc**	ICH	vc			Very Wet Cold		
1	**ICH vk 1**	ICH	vk	1		Very Wet Cool	Mica	
1	**ICH vk 1c**	ICH	vk	1	c	Very Wet Cool	Mica	Cold Air
1	**ICH vk 2**	ICH	vk	2		Very Wet Cool	Slim	
2	**ICH wc**	ICH	wc			Wet Cold		
1	**ICH wk 1**	ICH	wk	1		Wet Cool	Wells Gray	
1	**ICH wk 1c**	ICH	wk	1	c	Wet Cool	Wells Gray	Cold Air
1	**ICH wk 2**	ICH	wk	2		Wet Cool	Quesnel	
1	**ICH wk 3**	ICH	wk	3		Wet Cool	Goat	
1	**ICH wk 4**	ICH	wk	4		Wet Cool	Cariboo	
4	**ICH xw**	ICH	xw			Very Dry Warm		
4	**IDF dk 1**	IDF	dk	1		Dry Cool	Thompson	
4	**IDF dk 1a**	IDF	dk	1	a	Dry Cool	Thompson	Grassland

4	**IDF dk 1b**	IDF	dk	1	b	Dry Cool	Thompson	Steep South
4	**IDF dk 2**	IDF	dk	2		Dry Cool	Cascade	
4	**IDF dk 2b**	IDF	dk	2	b	Dry Cool	Cascade	Steep South
4	**IDF dk 3**	IDF	dk	3		Dry Cool	Fraser	
4	**IDF dk 4**	IDF	dk	4		Dry Cool	Chilcotin	
4	**IDF dm 1**	IDF	dm	1		Dry Mild	Kettle	
4	**IDF dm 2**	IDF	dm	2		Dry Mild	Kootenay	
4	**IDF dm 2a**	IDF	dm	2	a	Dry mild	Kootenay	Moist
4	**IDF dw**	IDF	dw			Dry Warm		
4	**IDF mw 1**	IDF	mw	1		Moist Warm	Okanagan	
4	**IDF mw 2**	IDF	mw	2		Moist Warm	Thompson	
4	**IDF mw 2a**	IDF	mw	2	a	Moist Warm	Thompson	Grassland
4	**IDF un**	IDF	un			Undefined		
4	**IDF ww**	IDF	ww			Wet Warm		
4	**IDF xh 1**	IDF	xh	1		Very Dry Hot	Okanagan	
4	**IDF xh 1a**	IDF	xh	1	a	Very Dry Hot	Okanagan	Grassland
4	**IDF xh 1b**	IDF	xh	1	b	Very Dry Hot	Okanagan	Steep South
4	**IDF xh 2**	IDF	xh	2		Very Dry Hot	Thompson	
4	**IDF xh 2a**	IDF	xh	2	a	Very Dry Hot	Thompson	Grassland
4	**IDF xh 2b**	IDF	xh	2	b	Very Dry Hot	Thompson	Steep South
4	**IDF xh 4**	IDF	xh	4		Very Dry Hot	Boundary	
4	**IDF xm**	IDF	xm			Very Dry Mild		
4	**IDF xw**	IDF	xw			Very Dry Warm		
4	**IDF xw a**	IDF	xw		a	Very Dry Warm		Grassland
4	**IDF xw b**	IDF	xw		b	Very Dry Warm		Steep South
1	**MH mm 1**	MH	mm	1		Moist Maritime	Windward	
1	**MH mm 2**	MH	mm	2		Moist Maritime	Leeward	
5	**MH mmp**	MH	mmp			Moist Maritime Parkland		
1	**MH un**	MH	un			Undefined		
1	**MH wh 1**	MH	wh	1		Wet Hypermaritime	Windward	
1	**MH wh 2**	MH	wh	2		Wet Hypermaritime	Leeward	
5	**MH whp**	MH	whp			Wet Hypermaritime Parkland		
3	**MS dc 1**	MS	dc	1		Dry Cold	Bridge	
3	**MS dc 2**	MS	dc	2		Dry Cold	Tatlayoko	
3	**MS dk**	MS	dk			Dry Cool		
3	**MS dm 1**	MS	dm	1		Dry Mild	Okanagan	
3	**MS dm 1a**	MS	dm	1	a	Dry Mild	Okanagan	Redcedar
3	**MS dm 2**	MS	dm	2		Dry Mild	Thompson	
3	**MS dv**	MS	dv			Dry Very Cold		
3	**MS un**	MS	un			Undefined		
3	**MS xk**	MS	xk			Very Dry Cool		
3	**MS xv**	MS	xv			Very Dry Very Cold		

4	**PP dh 1**	PP	dh	1		Dry Hot	Kettle	
4	**PP dh 2**	PP	dh	2		Dry Hot	Kootenay	
4	**PP xh 1**	PP	xh	1		Very Dry Hot	Okanagan	
4	**PP xh 1a**	PP	xh	1	a	Very Dry Hot	Okanagan	Grassland
4	**PP xh 2**	PP	xh	2		Very Dry Hot	Thompson	
4	**PP xh 2a**	PP	xh	2	a	Very Dry Hot	Thompson	Grassland
3	**SBPSdc**	SBPS	dc			Dry Cold		
3	**SBPSmc**	SBPS	mc			Moist Cold		
3	**SBPSmk**	SBPS	mk			Moist Cool		
3	**SBPSxc**	SBPS	xc			Very Dry Cold		
3	**SBS dh 1**	SBS	dh	1		Dry Hot	McLennan	
3	**SBS dh 2**	SBS	dh	2		Dry Hot	Robson	
3	**SBS dk**	SBS	dk			Dry Cool		
3	**SBS dw 1**	SBS	dw	1		Dry Warm	Horsefly	
3	**SBS dw 2**	SBS	dw	2		Dry Warm	Blackwater	
3	**SBS dw 3**	SBS	dw	3		Dry Warm	Stuart	
3	**SBS mc 1**	SBS	mc	1		Moist Cold	Moffat	
3	**SBS mc 2**	SBS	mc	2		Moist Cold	Babine	
3	**SBS mc 3**	SBS	mc	3		Moist Cold	Kluskus	
3	**SBS mh**	SBS	mh			Moist Hot		
3	**SBS mk 1**	SBS	mk	1		Moist Cool	Mossvale	
3	**SBS mk 2**	SBS	mk	2		Moist Cool	Williston	
3	**SBS mm**	SBS	mm			Moist Mild		
3	**SBS mw**	SBS	mw			Moist Warm		
3	**SBS un**	SBS	un			Undefined		
2	**SBS vk**	SBS	vk			Very Wet Cool		
2	**SBS wk 1**	SBS	wk	1		Wet Cool	Willow	
2	**SBS wk 2**	SBS	wk	2		Wet Cool	Finlay-Peace	
3	**SBS wk 3**	SBS	wk	3		Wet Cool	Takla	
3	**SBS wk 3a**	SBS	wk	3	a	Wet Cool	Takla	Douglas Fir
2	**SWB dk**	SWB	dk			Dry Cool		
2	**SWB mk**	SWB	mk			Moist Cool		
5	**SWB mks**	SWB	mks			Moist Cool Scrub		
2	**SWB un**	SWB	un			Undefined		
2	**SWB vk**	SWB	vk			Very Wet Cool		

Appendix E – Order Establishing Objectives for Old Growth Management Areas

The following pages contain the Order Establishing the Objectives for Old Growth Management Areas for the Marble Landscape Unit.

Preamble

This order establishes objectives for Old Growth Management Areas within three landscape units located on northern Vancouver Island.

The goal of these objectives is to contribute to biological diversity at the landscape level.

This preamble is intended to provide context and background; it does not, however, form part of the order.

PROVINCE OF BRITISH COLUMBIA

Ministry of Agriculture and Lands

Ministerial Order

Land Use Objectives for Old Growth Management Areas (OGMAs) within the Nahwitti, Tsulquate and Marble Landscape Units situated on northern Vancouver Island within the North Island – Central Coast Forest District.

Part 1 - Interpretation

1. Pursuant to Section 93.4 of the *Land Act,* the following objectives are established as land use objectives for the purposes of the *Forest and Range Practices Act* and apply to OGMAs within the Nahwitti, Tsulquate and Marble Landscape Units shown on the maps attached as Schedules 1 and 2.
2. Nothing in, under or arising out of this order either abrogates or derogates from any aboriginal rights, aboriginal title or treaty rights of any applicable First Nation, nor relieves the Province of any obligation to consult with any applicable First Nation.
3. Where an objective refers to an area shown on a map and the area is also defined by a spatial dataset, the area as defined by the spatial dataset will apply. All spatial datasets are available at http://www.geobc.gov.bc.ca.
4. In the event of any inconsistency between the location of an OGMA boundary as described in a spatial data set and the actual location as determined in the field, the latter shall apply.

Part 2 - Objectives

5. Objectives for Old Growth Management Areas

   (1) Maintain or recruit old growth forests in established Old Growth Management Areas, as shown on the map attached as Schedules 1 and 2, subject to subsections (2) to (6) below.

(2) Despite subsection 1, timber harvesting and road or bridge construction are permitted within OGMAs, provided that:

i) the area of the OGMA that is subject to timber harvesting or road or bridge construction does not exceed the greater of one hectare or 10% of the area of the OGMA,

ii) replacement forest is identified which is

i. biologically suitable

ii. of equivalent age, structure and area, and,

iii. situated in order of priority, either immediately adjacent to the OGMA, or adjacent to another OGMA in the same variant and landscape unit as the existing OGMA, and

iii) road or bridge construction, if applicable, is required to access resource values beyond or adjacent to the OGMA and no other practicable option for road or bridge location exists.

(3) For the purposes of subsection (2)(b) and (c), as an alternative to identifying replacement area, a temporary road or bridge site may be permanently deactivated and rehabilitated within four years after construction.

(4) Within OGMAs, the following activities are permitted:

a) First Nations traditional use of trees or understory plants.

b) Topping or pruning of trees along boundaries to improve wind firmness.

c) Timber harvesting to prevent the spread of insect infestations or diseases that pose significant threat to forested areas outside of OGMAs. Salvage within OGMAs will be done in a manner that retains as many old growth forest attributes as possible.

d) Road maintenance, deactivation, removal of danger trees, or brushing and clearing on existing roads under active tenure within the right-of-way for safety purposes.

e) Felling for guyline clearance, tailhold anchor trees, or danger trees along cutblock boundaries or within the right-of-way on new road/bridge alignments to meet safety requirements.

f) Silvicultural habitat enhancement in recruitment OGMAs.

g) Trail and site maintenance or development, and silvicultural treatments to address public safety associated with the management and operation of recreation sites and trails.

(5) Replacement forest is required if the total area of an OGMA that is subject to the activities pursuant to subsection (4) exceeds the greater of one hectare or 10% of the area of the OGMA. Replacement forest must be biologically suitable, of equivalent age, structure and area, and situated, in order of priority, either immediately adjacent to the OGMA, or adjacent to another OGMA in the same variant and landscape unit as the OGMA.

(6) OGMA adjustments and replacements made in accordance with subsections (2) to (5) must be documented and submitted as digital spatial data to the delegated decision maker at the end of each calendar year.

Part 3 - Effective Date and Transition

6. Application of this order

(1) This order and the land use objectives in this order take effect on the date that notice of this order is published in the Gazette.

(2) The period of time under section 8 of the *Forest and Range Practices Act* is two years.

H MacKnight

Heather MacKnight, MCIP
Regional Executive Director
Integrated Land Management Bureau
Coast Region

26 July 2010
Date

Appendix F Wildlife Notices and Orders

BRITISH COLUMBIA

March 2, 2006

NOTICE – INDICATORS OF THE AMOUNT, DISTRIBUTION AND ATTRIBUTES OF WILDLIFE HABITAT REQUIRED FOR THE SURVIVAL OF SPECIES AT RISK IN THE NORTH ISLAND – CENTRAL COAST FOREST DISTRICT

This Notice is given under the authority of section 7(2) of the *Forest Planning and Practices Regulation* (B.C. Reg. 14/04) and 9(3) of the *Woodlot Licence Planning and Practices Regulation* (B.C. Reg. 21/04).

The following Notice includes indicators of the amount, distribution and attributes of wildlife habitat required for the survival of the species at risk outlined in Schedule 1.

Approved Wildlife Habitat Areas are not included in the indicators of amount, distribution and attributes for each of the species outlined in Schedule 1. As per section 7(3) of the *Forest Planning and Practices Regulation*, forest tenure holders are exempt from the obligation to specify a result or strategy in relation to the objective set out in section 7(1) of the *Forest Planning and Practices Regulation*, for approved Wildlife Habitat Areas.

This Notice applies to the North Island – Central Coast Forest District.

This Notice replaces the Notice issued for the North Island – Central Coast Forest District on December 30, 2004.

Schedule 1

1) Coastal Tailed Frog (*Ascaphus truei*)

Amount:

1. 1549 ha not exceeding an impact to the mature timber harvesting landbase of 183 ha.

Distribution:

1. The amount of habitat referenced above must be distributed to provide:
   - areas of suitable habitat of the size, spatial distribution and connectivity identified in the species account for Coastal Tailed Frog the *Accounts and Measures for Managing Identified Wildlife* (Identified Wildlife Management Strategy Version 2004).
2. The areas described above are located within the biogeoclimatic units and preferred elevations identified in the species account for Coastal Tailed Frog in the *Accounts and Measures for Managing Identified Wildlife* in the Identified Wildlife Management Strategy Version 2004.

Attributes:

Species:
Coastal Tailed Frog

Attribute	Characteristics
Size	Approximately 20 ha (depending on number and length of suitable stream reaches). Larger areas may be appropriate in watersheds with unstable terrain (class 4-5). Areas should include at least two streams or stream reaches (i.e., S4 to S6) with previous detections of tailed frogs. The area should include a 30 m core area buffered by a 20m management zone on both sides of occupied stream reaches.
Habitat Attributes	Tailed frog aquatic habitats are generally characterised by year round flow, non fish bearing (S4-S6), intermediate gradient (>2.5%), coarse substrates (>6.4 cm), stable channel beds and forest cover (generally associated with structural stage S6 or S7). Retain 100% of forest cover within the core area. Within the management zone maintain 70% basal area with appropriate structure to maintain riparian forest, important structural elements (e.g., coarse wood debris,) water quality and temperature (5 to 18 degrees), and naturally dispersed water flows.
Elevation	From sea level to 2140 m.

2) Great Blue Heron (*Ardea herodias fannini*)

Amount:

1. 240 ha not exceeding an impact to the mature timber harvesting landbase of 36 ha.

Distribution:

1. The amount of habitat referenced above must be distributed to provide:
   - areas of suitable habitat of the size, spatial distribution and connectivity identified in the species account for Great Blue Heron in the *Accounts and Measures for Managing Identified Wildlife* (Identified Wildlife Management Strategy Version 2004).
2. The areas described above are located within the biogeoclimatic units and preferred elevations identified in the species account for Great Blue Heron in the *Accounts and Measures for Managing Identified Wildlife* in the Identified Wildlife Management Strategy Version 2004.

Attributes:

Species:
Great Blue Heron

Attribute	Characteristics
Size	Typically 80 ha to accommodate nesting areas/colonies, but may also target foraging areas. The core area should be approximately 12 ha and should include known nesting trees, foraging areas and flight paths. The boundary of the core area should be a 200 m radius around the aforementioned features. A management area should be captured to include a 300 m radius surrounding the core area.
Tree Features	Nests occur in fragmented forests, contiguous forest and isolated trees.
Tree Species	Red alder, black cottonwood, Douglas fir, Big-leaf maple, lodgepole pine, Sitka spruce, western white pine, hybrid white spruce, ponderosa pine, western red cedar, western hemlock.
Structural Stage	5 (young forest), 6 (mature forest), 7(old forest).
Elevation	Sea level to 1100 m.

3) Marbled Murrelet (*Brachyramphus marmoratus*)

Amount:

1. An amount equal to the total amount of currently suitable nesting habitat in the non-contributing landbase. Government policy for determining the amount of suitable nesting habitat is provided in the species account for Marbled Murrelet in the *Accounts and Measures for Managing Identified Wildlife* (Identified Wildlife Management Strategy Version 2004);
2. Within Tree Farm License 37, a total of 7,100 ha of suitable nesting habitat within the non-contributing landbase;
3. An amount of suitable Marbled Murrelet nesting habitat within Old Growth Management Areas consistent with the direction from landscape unit planning; and
4. An amount of suitable nesting habitat to a maximum net mature timber harvesting landbase impact of 1434 ha.

Distribution:

1. The amount of habitat referenced above must be distributed to provide:
   - areas of suitable habitat of the size, spatial distribution and connectivity identified in the species account for Marbled Murrelet in the *Accounts and Measures for Managing Identified Wildlife* (Identified Wildlife Management Strategy Version 2004).
2. The areas described above are located within the biogeoclimatic units and preferred elevations identified in the species account for Marbled Murrelet in the *Accounts and Measures for Managing Identified Wildlife* in the Identified Wildlife Management Strategy Version 2004.

Attributes:

Species:
Marbled Murrelet

Attribute	Characteristics
Size	Maintain a balanced range of patch sizes including a mix of large (>200 ha), medium (50-200 ha) and small (<50 ha) patches within managed forests. The area should include vertical canopy complexity,
Tree Features	Large branches or branches with deformities, and presence of mossy platforms
Tree Species	It is unlikely that Marbled murrelets select particular tree species, however certain species are more likely to provide large horizontal platforms suitable for nesting. This includes yellow cedar, western hemlock, Sitka spruce, Douglas-fir and western red cedar. Less likely species include mountain hemlock and amabilis fir.
Nesting Habitat Features	Suitable nesting habitat includes old seral stage coniferous forests, providing large trees with platforms (limbs or deformities >15cm diameter) with variable canopy structure and small gaps in the canopy. Readily nest on steep slopes but is not essential if forest canopies are non-uniform.
Tree Size	Most nesting trees in BC are >200 yr. Nest trees are typically >40 m tall and nest heights are typically >30 m. Nest limbs range in size from 15-74 cm diameter.
Structural Stage	7: old forest (>250 yr - age class 9, but 8 is acceptable if older forest is not present and the age class 8 provides platform limbs and other nest attributes).
Additional information	Table 3 of the IWMS Version 2004 species account for Marbled Murrelet provides detailed information about the habitat features that are associated with most likely, moderately likely and least likely habitat within each of the Marbled Murrelet Conservation regions.

4) "Queen Charlotte" Goshawk (*Accipiter gentilis laingi*)

Amount:

1. 277 ha not exceeding an impact to the mature timber harvesting landbase of 128 ha.

Distribution:

1. The amount of habitat referenced above must be distributed to provide:
   - areas of suitable habitat of the size, spatial distribution and connectivity identified in the species account for "Queen Charlotte" Goshawk in the *Accounts and Measures for Managing Identified Wildlife* (Identified Wildlife Management Strategy Version 2004).
   - areas of suitable goshawk breeding habitat to minimize overlap between goshawk home ranges (approximately 5-8 km separation).
2. The areas described above are located within the biogeoclimatic units and preferred elevations identified in the species account for "Queen Charlotte" Goshawk in the *Accounts and Measures for Managing Identified Wildlife* in the Identified Wildlife Management Strategy Version 2004.

Attributes:

Species:
Queen Charlotte Goshawk

Attribute	Characteristics
Nesting and PFA Area	Approximately 200 ha in size.
Stand Features (Nesting and PFA)	Structural stage 5-7 (>45yrs), multi-layered canopies, structurally diverse, canopy closure (greater than 50%), large diameter trees (for the locale), snags and course woody debris (CWD), typically not along forest/non-forest edges. Not near urban areas and generally on the lower 2/3 of slopes where slope gradient is <40%.
Tree Species (Nesting and PFA)	Western hemlock, Douglas fir, Sitka spruce, Western red cedar, amabilis fir and red alder.
Structural Stage	5 (young forest - is used but is generally not preferred), 6 (mature forest) and 7 (old forest).
Elevation (Nesting and PFA)	Areas managed for nesting must generally be below 900 m.

5) Grizzly Bear (*Ursus arctos*)

Amount:

1. Within the Mid Coast TSA, 16,000 ha not exceeding an impact to the timber harvesting landbase of 6,046 ha;
2. Within TFL 39 Block 7, 1320 ha not exceeding an impact to the timber harvesting land base of 126 ha; and
3. Within the Mid Coast TSA, 1700 ha not exceeding an impact to the mature timber harvesting landbase of 170 ha.

Distribution:

1. The amount of habitat referenced above must be distributed to provide:

a) Within the Mid Coast TSA, areas of suitable habitat of the size, spatial distribution and connectivity consistent with the distribution of Grizzly Bear polygons as per TSR 1 and 2.

b) Within TFL 39 Block 7, areas of suitable habitat of the size, spatial distribution and connectivity consistent with the distribution of Grizzly Bear polygons as per Management and Working Plan #8.

c) Within the Mid Coast TSA:
   - areas of suitable habitat of the size, spatial distribution and connectivity. identified in the species account for Grizzly Bear in the *Accounts and Measures for Managing Identified Wildlife* (Identified Wildlife Management Strategy Version 2004).
   - The areas described above are located within the biogeoclimatic units and preferred elevations identified in the species account for Grizzly Bear in the

Accounts and Measures for Managing Identified Wildlife in the Identified Wildlife Management Strategy Version 2004.

Attributes:

a) Within the Mid Coast TSA the amount referenced above should include attributes consistent with those areas identified for Grizzly Bear habitat identified in TSR 1 and 2;
b) Within TFL 39 Block 7 the amount referenced above should include attributes consistent with those areas identified for Grizzly Bear habitat identified in management and Working Plan #8; and
c) Within the Mid Coast TSA, attributes consistent with those identified below:

Species:
Grizzly Bear

Attribute	Characteristics
Size	1-500 ha, depending on the area of use, extent of seasonal habitat and buffer size required.
Critical patch habitats	Critical patch habitats include beaches and beach margins, estuaries, rich non-forested fens, the edges of forested and non-forested bogs, herb-dominated patches on avalanche chutes with adjacent forest (particularly south-facing ones), herb-dominated subalpine parkland meadows, skunk cabbage swamps, floodplain ecosystems, and areas where bears fish for spawning salmon. Den cavities and surrounding stands are also considered critical. Non-forested critical habitats include a core area and buffer of forested cover. Forested critical habitats are not buffered.
Denning Habitat Features	Hibernating habitats tend to be high elevation areas that are sloped with dry, stable soil conditions that remain frozen throughout the winter. Dens are typically located on steep north-facing slopes, areas where vegetation will stabilize the den roof and where snow will accumulate for insulation. Dens are rarely re-used but Grizzly bears will often return to the same vicinity to dig new dens.
Foraging Habitat Features	Habitat selection is strongly influenced by meeting nutritional requirements, access to mates, thermal cover (i.e., dens), social interactions and the presence and activities of people. Habitat requirement vary greatly as some bears are more transient while others are more resident. Both residents and transients select patches or complexes of habitats within landscapes.
Structural Stage	Generally, foraging is more abundant in non-forested sites, sites with partial forest or sites with many tree gaps in older forest. Closed forest sites near quality habitat may be used for security and day bedding areas. Many or all structural stages can be used seasonally or for specific needs and as such, forage type is not necessarily tied to one particular structural stage.
Elevation	All elevations from sea level estuaries to high alpine meadows and talus slopes.

Signed this 2 day of March, 2006
Rod Davis, Director Ecosystems Branch
Ministry of Environment

BRITISH COLUMBIA

ORDER – Wildlife Habitat Areas # 1-184, 1-185, 1-204, 1-205

This order is given under the authority of sections 9(2) and 10(1) of the *Government Actions Regulation* (B.C. Reg. 582/04).

The Deputy Minister of Environment orders that:

1. the wildlife habitat areas shown in the map set out in the attached Schedule As (1-184, 1-185, 1-204 & 1-205) and boundaries contained in the GIS file *twha_bc* are established;
2. the wildlife habitat areas in the attached Schedule As and boundaries contained in the GIS file *twha_bc* are established for Marbled Murrelet (*Brachyramphus marmoratus*);
3. the general wildlife measures outlined in Schedule 1 are established for the wildlife habitat areas in the attached Schedule As and boundaries contained in the GIS file *twha_bc*;
4. for the purposes of section 2(3)(a) of the *Government Actions Regulation*, these General Wildlife Measures apply to minor tenures; and
5. where there is discrepancy between the wildlife habitat area boundaries as shown in the attached Schedule As and the GIS file *twha_bc*, the boundaries as detailed in the GIS file will take precedent. The centre point of the line on the map denoting the wildlife habitat area is what establishes the boundary.
6. pursuant to section 7(3) of the Forest Planning and Practices Regulation the person(s) required to prepare a forest stewardship plan are hereby exempted from the obligation to prepare results or strategies in relation to the objective set out in section 7(1) of the Forest Planning and Practices Regulation to the extent that wildlife habitat areas 1-184, 1-185, 1-204 and 1-205 address the amount included for Marbled Murrelet in the Notice for the North Island-Central Coast Forest District.

Schedule 1 – General Wildlife Measures

Access

· Do not construct or widen roads unless the Minister of Environment or delegate is satisfied there is no other practicable option and approves an exemption.

Harvesting and Silviculture

· Do not harvest timber except for salvage or traditional aboriginal purposes when the Minister of Environment or delegate approves an exemption.

Pesticides

· Do not use pesticides unless an exemption is approved by the Minister of Environment or delegate.

Recreation

· Do not develop recreational structures, trails or facilities.

Signed this 27 day of Sept, 2006
Chris Trumpy, Deputy Minister
Ministry of Environment

GOVERNMENT ACTIONS REGULATION ORDER
WILDLIFE HABITAT AREAS
Marbled Murrelets

North Island Central Coast Natural Resource District Wildlife Habitat Areas:

Table 1:

1-280	1-304	1-327	1-626
1-282	1-308	1-328	1-627
1-290	1-311	1-332	1-628
1-292	1-312	1-359	1-630
1-294	1-313	1-380	1-690a,b
1-295	1-315	1-385	1-691

This order is given under the authority of Sections 9(2) and 10(1) of the Government Actions Regulation (GAR) (B.C. Reg. 582/2004).

1. The Assistant Deputy Minister, Coast Area, Ministry of Forests, Lands, Natural Resource Operations and Rural Development, being satisfied that:
   i. the following area contains habitat that is necessary to meet the habitat requirements for Marbled Murrelets (*Brachyramphus marmoratus*); and
   ii. the habitat requires special management that is not otherwise provided for under GAR or another enactment;

orders that

a) the areas shown in the map set out in the attached Schedule A (Table 1) and contained in the wildlife habitat area (WHA) spatial layer stored in the Geographic Warehouse (WHSE_WILDLIFE_MANAGEMENT.WCP_WILDLIFE_HABITAT_ AREA_POLY*)* are established as wildlife habitat areas WHAs (see Table 1 above) for Marbled Murrelets. The centre point of the line on the attached Schedule A is what establishes the WHA boundary; and
b) if there is a discrepancy between the areas shown in the map set out in the attached Schedule A and the WHA spatial layer stored in the Geographic Warehouse (WHSE_WILDLIFE_MANAGEMENT.WCP_WILDLIFE_HABITAT_AREA_POLY), the areas as detailed in the WHA spatial layer will take precedent; and
c) pursuant to Section 7(3) of the Forest Planning and Practices Regulation the person(s) required to prepare a forest stewardship plan are hereby exempted from the obligation to prepare results or strategies in relation to the objective set out in Section 7(1) of the Forest Planning and Practices Regulation to the extent that WHAs (see Table 1 above) address the amount included for Marbled Murrelet in the notices for the North Island Central Coast Natural Resource District.

2. The Assistant Deputy Minister, Coast Area, Ministry of Forests, Lands, Natural Resource Operations and Rural Development, being satisfied that

i. the general wildlife measures (GWMs) described below are necessary to protect or conserve Marbled Murrelets and the habitat of Marbled Murrelets; and

ii. GAR or another enactment does not otherwise provide for that protection or conservation;

orders that

a) the GWMs outlined in Schedule 1 are established for the 24 WHAs listed in Table 1:

Definitions

Words and expressions not defined in this order have the meaning given to them in the *Forest and Range Practices Act* and the regulations made under it, unless context indicates otherwise.

Schedule 1 – General Wildlife Measures:

Access:

1. Do not construct roads.
2. GWM 1 does not apply to WHA-1-628 where a switchback road may be constructed from the northwest side of the WHA to the south side in order to access timber upslope of the WHA.
3. GWM 1 does not apply to WHA-1-295 where a road MC10500 may infringe slightly on the northern end of the WHA.

Harvesting and Silviculture:

4. Do not fell trees or conduct timber harvesting, salvage or silvicultural activities, including crown modifications or windfirming treatments.
5. GWM 2 does not apply for felling of danger trees for worker safety and where guyline anchors and tailholds are required to facilitate timber harvesting outside a WHA.
6. Trees felled in accordance with GWM 3 must be retained on-site to provide coarse woody debris.

Signed this 23rd day of March, 2018.

Craig Sutherland
Assistant Deputy Minister, Coast Area
Ministry of Forests, Lands, Natural Resource Operations and Rural Development

Appendix 1:

The following information is provided as background information and support to the order establishing 24 wildlife habitat areas (WHA) listed in Table 1. This appendix is not part of the order.

1. Activities to which the order does not apply: Section 2(2) of the Government Actions Regulation states

   An order under any of Sections 5 to 15 does not apply in respect of

   (a) any of the following entered into before the order takes effect:

   (i) a cutting permit;

   (ii) a road permit;

   (iii) a timber sale licence that does not provide for cutting permits;

   (iv) a forestry licence to cut issued by a timber sales manager under Section 47.6 (3) of the *Forest Act*;

   (v) subject to Subsection (3), a minor tenure,

   (b) a declared area,

   (c) areas described in Section 196 (1) of the Act, and

   (d) areas referred to in Section 110 of the Forest Planning and Practices Regulation.

2. Authority to consider an exemption from these general wildlife measures is provided in Section 92(1) of the Forest Planning and Practices Regulation, Section 79(1) of the Woodlot Licenses Planning and Practices Regulation and Section 36(3) of the Range Planning and Practices Regulation. An exemption may be provided if the Minister's delegate is satisfied that the intent of the general wildlife measure will be achieved or that compliance with the provision is not practicable, given the circumstances or conditions applicable to a particular area.

   An exemption application should be submitted to the Minister's delegate (FLNRORD Director of Resource Management) for the region that the WHA is located with a rationale describing the nature of the problem and options to integrate WHA conservation with proposed forest practices. This submission will assist in timely consideration of the matter, and will inform the conditions, if any, of the exemption that may be granted prior to commencement of activities. Upon receipt of a complete exemption application, a determination will normally be made within 14 days of arrival. Incomplete packages will be returned to the proponent for re-submission.

GOVERNMENT ACTIONS REGULATION ORDER
WILDLIFE HABITAT AREAS
Marbled Murrelets

Table 1 - Campbell River Natural Resource District Wildlife Habitat Areas:

1-232a	1-499	1-505	1-512	1-518	1-525	1-548	1-564	1-640	1-694
1-232b	1-500	1-506	1-513	1-519	1-526a	1-555	1-566	1-641	
1-475	1-501	1-507	1-514	1-521	1-528	1-559	1-580	1-642	
1-477	1-502	1-509	1-515	1-522	1-529	1-560	1-582	1-643	
1-497	1-503	1-510	1-516	1-523	1-539	1-562	1-584	1-644	
1-498	1-504	1-511	1-517	1-524	1-547	1-563	1-639	1-645	

Table 2 - North Island Central Coast Natural Resource District Wildlife Habitat Areas:

1-591	1-606	1-614	1-620		1-661	1-671	1-678
1-595	1-607	1-615	1-622	1-654	1-662	1-672	1-679
1-597	1-609	1-616	1-623	1-655	1-663	1-673	1-680
1-599	1-611	1-617	1-624	1-657	1-666	1-674	1-681
1-602	1-612	1-618	1-648	1-658	1-668	1-676	1-682
1-604	1-613	1-619	1-649	1-659	1-670	1-677	

This order is given under the authority of Sections 9(2) and 10(1) of the Government Actions Regulation (GAR) (B.C. Reg. 582/2004).

1. The Assistant Deputy Minister, Coast Area, Ministry of Forests, Lands, Natural Resource Operations and Rural Development, being satisfied that:
   i. the following area contains habitat that is necessary to meet the habitat requirements for Marbled Murrelets (*Brachyramphus marmoratus*); and
   ii. the habitat requires special management that is not otherwise provided for under GAR or another enactment;

   orders that:
   a) the areas shown in the map set out in the attached Schedule A (Tables 1 and 2 above) and contained in the wildlife habitat area (WHA) spatial layer stored in the Geographic Warehouse (WHSE_WILDLIFE_MANAGEMENT.WCP_WILDLIFE_HABITAT_AREA_POLY) are established as wildlife habitat areas WHAs (see Tables 1 and 2 above) for Marbled Murrelets. The centre point of the line on the attached Schedule A is what establishes the WHA boundary; and
   b) if there is a discrepancy between the areas shown in the map set out in the attached Schedule As and the WHA spatial layer stored in the Geographic Warehouse (WHSE_WILDLIFE_MANAGEMENT.WCP_WILDLIFE_HABITAT_AREA_POLY), the areas as detailed in the WHA spatial layer will take precedent; and

c) pursuant to Section 7(3) of the Forest Planning and Practices Regulation the person(s) required to prepare a forest stewardship plan are hereby exempted from the obligation to prepare results or strategies in relation to the objective set out in Section 7(1) of the Forest Planning and Practices Regulation to the extent that wildlife habitat areas (see Tables 1 and 2) address the amount included for Marbled Murrelet in the Notices for the North Island Central Coast and Campbell River Natural Resource Districts.

2. The Assistant Deputy Minister, Coast Area, Ministry of Forests, Lands, Natural Resource Operations and Rural Development, being satisfied that:

i. the general wildlife measures (GWMs) described below are necessary to protect or conserve Marbled Murrelets and the habitat of Marbled Murrelets; and

ii. GAR or another enactment does not otherwise provide for that protection or conservation;

orders that:

a) the GWMs outlined in Schedule 1 are established for the 101 WHAs listed in Tables 1 and 2:

Definitions

Words and expressions not defined in this order have the meaning given to them in the *Forest and Range Practices Act* and the regulations made under it, unless context indicates otherwise.

Schedule 1 – General Wildlife Measures:

Access:

1. Do not construct roads.

Harvesting and Silviculture:

2. Do not fell trees or conduct timber harvesting, salvage or silvicultural activities, including crown modifications or windfirming treatments.
3. GWM 2 does not apply for felling of danger trees for worker safety and where guyline anchors and tailholds are required to facilitate timber harvesting outside a WHA.
4. Trees felled in accordance with GWM 3 must be retained on-site to provide coarse woody debris.

Signed this 23rd day of March, 2018.

Craig Sutherland
Assistant Deputy Minister, Coast Area
Ministry of Forests, Lands, Natural Resource Operations and Rural Development

Appendix 1:

The following information is provided as background information and support to the order establishing 101 WHAs listed in Tables 1 and 2. This appendix is not part of the order.

1. Activities to which the order does not apply: Section 2(2) of the Government Actions Regulation states

   An order under any of Sections 5 to 15 does not apply in respect of

   (a) any of the following entered into before the order takes effect:

   (i) a cutting permit;

   (ii) a road permit;

   (iii) a timber sale licence that does not provide for cutting permits;

   (iv) a forestry licence to cut issued by a timber sales manager under Section 47.6 (3) of the *Forest Act*;

   (v) subject to subsection (3), a minor tenure,

   (b) a declared area,

   (c) areas described in Section 196 (1) of the Act, and

   (d) areas referred to in Section 110 of the Forest Planning and Practices Regulation.

2. Authority to consider an exemption from these GWMs is provided in Section 92(1) of the Forest Planning and Practices Regulation, Section 79(1) of the Woodlot Licenses Planning and Practices Regulation and Section 36(3) of the Range Planning and Practices Regulation. An exemption may be provided if the Minister's delegate is satisfied that the intent of the GWM will be achieved or that compliance with the provision is not practicable, given the circumstances or conditions applicable to a particular area.

   An exemption application should be submitted to the Minister's delegate (FLNRORD Director of Resource Management) for the region that the WHA is located with a rationale describing the nature of the problem and options to integrate WHA conservation with proposed forest practices. This submission will assist in timely consideration of the matter, and will inform the conditions, if any, of the exemption that may be granted prior to commencement of activities. Upon receipt of a complete exemption application, a determination will normally be made within 14 days of arrival. Incomplete packages will be returned to the proponent for re-submission.

BRITISH
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ORDER – Wildlife Habitat Areas
 # 1-214 and 1-215
Red-legged Frog –North-Island Central Coast Forest District

This order is given under the authority of sections 9(2) and 10(1) of the *Government Actions Regulation* (B.C. Reg. 582/2004) (GAR).

1. The Deputy Minister of Environment, being satisfied that
   i. the following areas contain habitat that is necessary to meet the habitat requirements for red-legged frog (*Rana aurora aurora*); and
   ii. the habitat requires special management that is not otherwise provided for under GAR or another enactment;

   orders that
   a) the areas shown in the map set out in the attached Schedule As (1-214 and 1-215) and contained in the wildlife habitat area (WHA) spatial layer stored in the Land and Resource Data Warehouse (*twha_bc)* are established as wildlife habitat areas 1-214 and 1-215 for red-legged frog. The centre point of the line on the attached Schedule As is what establishes the WHA boundary; and
   b) if there is a discrepancy between the areas shown in the map set out in the attached Schedule As and the WHA spatial layer stored in the Land and Resource Data Warehouse (*twha_bc*), the areas as detailed in the WHA spatial layer will take precedent.

2. The Deputy Minister of Environment, being satisfied that
   i. the general wildlife measures (GWMs) described below are necessary to protect and conserve red-legged frog; and
   ii. GAR or another enactment does not otherwise provide for that protection or conservation;

   orders that
   a) the GWMs outlined in Schedule 1 are established for WHAs 1-214 and 1-215.

Definitions

Words and expressions not defined in this order have the meaning given to them in the *Forest and Range Practices Act* (FRPA) and the regulations made there under, unless context indicates otherwise.

Schedule 1 – General Wildlife Measures
1-214

Access

1. Do not construct roads.

Harvesting and silviculture

2. Do not conduct timber harvesting, except as provided in GWM 3.
3. GWM 2 does not apply where:
   a) guyline tiebacks are required to facilitate timber harvesting; or,
   b) trees are felled in accordance with section 2(3) of the *Forest Planning and Practices Regulation* (FPPR); and,
   c) trees felled for the purposes in (a) and (b) that fall within the WHA are retained on-site.

Pesticides

4. Do not use pesticides, except as provided in GWM 5.
5. GWM 5 does not apply for the application of herbicides to control for invasive plants or noxious weeds, if the herbicide is applied:
   a) by stem injection, cut and paint, foliar wipe or other direct plant application; or
   b) by spot spraying for individual plants or cluster of plants if direct plant application is not practicable; and
   c) with no direct application to either red-legged frogs or water.

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Access

6. Do not construct roads.

Harvesting and silviculture

7. Do not conduct timber harvesting, except as provided in GWM 8.
8. GWM 7 does not apply where:
   a) windfirming and crown modification treatments are required in the management zone;
   b) guyline tiebacks are required to facilitate timber harvesting; or,
   c) trees are felled in accordance with section 2(3) of the *Forest Planning and Practices Regulation* (FPPR); and,
   d) trees felled for the purposes in (a) and (b) that fall within the WHA are retained on-site.

Pesticides

9. Do not use pesticides, except as provided in GWM 10.
10. GWM 9 does not apply for the application of herbicides to control for invasive plants or noxious weeds, if the herbicide is applied:
    a) by stem injection, cut and paint, foliar wipe or other direct plant application; or
    b) by spot spraying for individual plants or cluster of plants if direct plant application is not practicable; and
    c) with no direct application to either red-legged frogs or water.

Signed this 1 day of April, 2009
Doug Konkin, Deputy Minister
Ministry of Environment

Appendix 1:

The following information is provided by MoE as background information and support to the order establishing these WHAs. This appendix is not part of the order.

1. **Activities to which the order does not apply:** Section 2(2) of the *Government Actions Regulation* states

   An order under any of sections 5 to 15 does not apply in respect of

   (a) any of the following entered into before the order takes effect:

   (i) a cutting permit;

   (ii) a road permit;

   (iii) a timber sale licence that does not provide for cutting permits;

   (iv) a forestry licence to cut issued by a timber sales manager under section 47.6 (3) of the Forest Act;

   (v) subject to subsection (3), a minor tenure,

   (b) a declared area,

   (c) areas described in section 196 (1) of the Act, and

   (d) areas referred to in section 110 of the Forest Planning and Practices Regulation.

2. Authority to consider an exemption from these GWMs is provided in section 92(1) of the FPPR. An exemption may be provided if the Minister's delegate is satisfied that the intent of the general wildlife measure will be achieved or that compliance with the provision is not practicable, given the circumstances or conditions applicable to a particular area.

   An exemption application should be submitted to the Minister's delegate (Regional Manager – MoE, for the Region that the WHA is located) with a rationale describing the nature of the problem and options to integrate WHA conservation with proposed forest practices. This submission will assist in timely consideration of the matter, and will inform the conditions, if any, of the exemption that may be granted prior to commencement of activities. Upon receipt of a complete exemption application, a determination will normally be made within 14 days of arrival. Incomplete packages will be returned to the proponent for re-submission.

3. An authorized person who carries out a primary forest activity should conduct the activity at a time and in a manner that is unlikely to create road-generated sedimentation that would harm red-legged frogs or destroy, damage or harmfully alter red-legged frog breeding habitat in the WHA. Management to consider:
   - Consider installing culverts under roads with drift fences directed toward the culverts at selected locations on roads that have high traffic volume at night and where road mortality of red-legged frogs is high.
   - Prevent fish introduction and the spread of bullfrogs.
   - To minimize the potential impacts of windthrow on the WHAs, when designing a cutblock adjacent to a WHA, conduct a windthrow risk

assessment and determine the appropriate course of action to protect the integrity of the WHA.

- To minimize the potential impacts of landslides and erosion, WHAs below steep or erodible ground should be assessed within this context. An assessment of landslide hazard from nearby slopes should consider the potential impact to a WHA and reduce the hazard accordingly.
- To minimize the potential impacts of roads:
  - minimize road densities and stream crossings;
  - plan the location of roads to avoid red-legged frog habitat elements and maintain the natural hydrology of water bodies through appropriate drainage management;
  - maintain non-erodible road surfaces and minimize road-generated sediment; and
  - employ sediment management techniques to minimize the introduction of sediment into streams and wetlands.
  - Any road construction that could affect the waterflow into the WHA should be completed between August and January which is outside of the breeding and larval development season (February through July).

BRITISH COLUMBIA

ORDER

AMENDMENT FOR UNGULATE WINTER RANGE #U-1-006 UNIT 2

This order is given under the authority of sections 9(2) and 12(1) of the Government Actions Regulation (B.C. Reg. 582/2004).

The Director of Resource Management of the West Coast Region, Ministry of Forests, Lands and Natural Resource Operations orders that:

1. The boundary of Unit 2 in Ungulate Winter Range U-1-006 is amended, as shown on the attached map "Amendment to UWR U-1-006 Polygon Unit No. 2" and contained in the UWR spatial layer stored in the GeoBC Geographic Warehouse (WHSE_WILDLIFE_MANAGEMENT.WCP_UNGULATE_WINTER_RANGE_SP). The centre point of the line on the map denoting the UWR is what establishes the boundary;
2. The boundary replaces the boundary approved for Unit 2 on October 18, 2004, for Black-tailed Deer (*Odocoileus hemionus columbianus*) and Roosevelt Elk (*Cervus elaphus roosevelti*);
3. Where there is any discrepancy between the UWR boundaries as shown in the attached map and the UWR spatial layer stored in the GeoBC Geographic Warehouse, the boundaries as detailed in the UWR spatial file will take precedent; and,
4. The general wildlife measures established for Ungulate Winter Range U-1-006 on October 18, 2004, continue to apply.

Signed this 4 day of Sept., 2014

Larry Barr P. Ag.
Director, Resource Management
West Coast Region
Ministry of Forests, Lands and Natural Resource Operations

Attachment: Amendment to UWR U-1-006 Polygon Unit No. 2 Map

BRITISH
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ORDER – AMENDMENT TO UNGULATE WINTER RANGE U-1-006, UNIT 3

This order is given under the authority of sections 9(2) and 12(1) of the *Government Actions Regulation* (B.C. Reg. 582/2004).

The Deputy Minister of the Ministry of Environment orders that:

1. the boundary of unit 3 in Ungulate Winter Range (UWR) U-1-006 is amended as shown on the attached map "Amendment to UWR U-1-006, Polygon Unit No. 3", and contained in the UWR spatial layer stored in the Land and Resource Data Warehouse (*tuwra_bc*). The centre point of the line on the map denoting the UWR is what establishes the boundary;
2. this boundary replaces the boundary approved for unit 3 on 18 October 2004, for black-tailed deer (*Odocoileus hemionus columbianus*) and Roosevelt elk (*Cervus elaphus roosevelti*);
3. where there is any discrepancy between the UWR boundaries as shown in the attached map and the UWR spatial layer stored in the Land and Resource Data Warehouse (*tuwra_bc*), the boundaries as detailed in the UWR spatial file will take precedent; and,
4. the general wildlife measures established for UWR U-1-006 on 18 October 2004 apply to the amended boundaries of unit 3.

Signed this 17 day of March, 2009
Doug Konkin, Deputy Minister
Ministry of Environment

BRITISH COLUMBIA

AMENDED ORDER – UNGULATE WINTER RANGE #U1-010

TFL 6

This order is given under the authority of sections 9(2) and 12(1) of the *Government Actions Regulation* (B.C. Reg. 582/2004).

The Deputy Minister of the Environment orders that:

1. this order replaces the order established on November 18, 2004 titled "ORDER – UNGULATE WINDTER RANGE #U1-010";
2. the ungulate winter range shown in the map set out in the Schedule A (#U1-010) established on November 18, 2004 is maintained;
3. the ungulate winter range is established for black-tailed deer (*Odocoileus hemionus columbianus*) and Roosevelt elk (*Cervus elaphus roosevelti*); and
4. the general wildlife measures outlined in Schedule 1 are established for the ungulate winter range in the Schedule A (#U1-010) established on November 18, 2004.

Schedule 1 – General Wildlife Measures

1. Road construction is not to occur within the designated ungulate winter ranges unless there is no other practicable option and an exemption is approved by the MOE delegated decision maker.

2. An exemption is not required for road maintenance, road deactivation, felling of danger trees or brushing and clearing on existing roads within the UWR. These activities will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat within the designated ungulate winter range.

3. Harvesting is not to occur within the designated ungulate winter ranges unless an exemption is approved by the MOE delegated decision maker. An exemption would be considered for the purposes of enhancing the quality of the winter range.

4. An exemption is not required for harvesting within the designated ungulate winter ranges when it is required to address worker safety: felling of danger trees, felling for guy line anchors, felling of tail hold anchor trees within an UWR along adjacent cutblock boundaries. Harvesting will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat within the designated ungulate winter ranges.

5. Trees that must be felled within an UWR will be left onsite to provide coarse woody debris, unless the felled tree lies outside the UWR.

6. Salvage harvesting is not to occur within the designated ungulate winter ranges unless an exemption is approved by the MOE delegated decision maker.

Note: It is intended that UWR #5 will be overlapped with an OGMA upon completion of Landscape Unit Planning and that the THLB impact associated with UWR #5 will be addressed through the landscape unit planning budget associated with OGMA establishment. Until the landscape unit planning process is completed the THLB impacts of ungulate winter range #5 in TFL 6 will be addressed through the THLB budget allocated to other wildlife values in TFL 6 (as per MP#9, TSR2). Upon completion of Landscape Unit Planning MWLAP in partnership with WFP will address the budget implications of UWR #5.

Signed this 16th day of May, 2007
Chris Trumpy, Deputy Minister

BRITISH COLUMBIA

AMENDED ORDER – UNGULATE WINTER RANGE #U1-011

Kingcome TSA

This order is given under the authority of sections 9(1), 9(2) and 12(1) of the *Government Actions Regulation* (B.C. Reg. 582/2004).

The Deputy Minister of Environment orders that:

1) the order established on June 09, 2006 titled "ORDER – UNGULATE WINTER RANGE #U1-011" be replaced by this order;

2) (a) the ungulate winter range polygon unit number WAK-04 shown in the map set out in the attached Schedule A (Amendment to UWR U-1-011 Polygon Unit No. WAK-04), replace unit number WAK-04 established on June 09, 2006, and is established;

   (b) the remaining ungulate winter range polygons shown in the map set out in the Schedule A (#U-1-011) established on June 09, 2006 are maintained;

3) the ungulate winter range is established for black-tailed deer (*Odocoileus hemionus columbianus*), Roosevelt elk (*Cervus elaphus roosevelti*), Rocky mountain goat (Oreamnus *americanus*), and Moose (*Alces alces*);

4) the general wildlife measures outlined in Schedule 1 are established for the ungulate winter range in the attached Schedule A;

5) the specified area as outlined in general wildlife measure 7, Schedule 1, is established;

6) for the purpose of section 2(3)(a) of the Government Actions Regulation, the general wildlife measures outlined in Schedule 1 apply to minor tenures;

7) where there is any discrepancy between the ungulate winter range boundaries as shown in the attached Schedules A and the GIS file *tuwra_bc*, the boundaries as detailed in the GIS file will take precedent. The centre point of the line on the map denoting the ungulate winter range is what establishes the boundary; and

8) pursuant to section 7(3) of the *Forest Planning and Practices Regulation* the person(s) required to prepare a forest stewardship plan are hereby exempted from the obligation to prepare results or strategies in relation to the objective set out in section 7(1) of the *Forest Planning and Practices Regulation* for the winter survival of ungulates in the Kingcome Timber Supply Area; and

9) the general wildlife measures outlined in schedule 1 do not apply for the purposes of exploration, development and production activities when these activities have been authorized for the purpose of subsurface resource exploration, development or production by the *Mineral Tenure Act, the Coal Act, the Mines Act, the Petroleum and Natural Gas Act, the Pipeline Act or the Geothermal Resources Act.*

Schedule 1: General Wildlife Measures:

1. Road construction is not to occur within the designated ungulate winder ranges unless there is no other practicable option and an exemption is approved by the MOE designated authority.

   An exemption is not required for the construction of future extension of road P10 in UWR Unit No. NAH- 07,the Nahwitti East Mainline road in the upland portion of UWR Unit No. NAH- 03 north of Nahwitti River and Basin Mainline road in Unit No. KLA-03. The extension of these roads will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat.

   MOE recognizes that the lower boundary of the UWR Unit No. WAK-08 is proposed along the 620 metre above sea level elevation line in the northwest portion of this UWR. It is expected that road construction would occur below this elevation. Amendments to the UWR boundary will be considered if it is not practicable to construct the road below this elevation.

2. An exemption is not required for road maintenance, road deactivation, felling of danger trees or brushing and clearing on existing roads within the Ungulate Winter Range. These activities will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat within the designated ungulate winter range.

3. Harvesting is not to occur within the designated ungulate winter ranges unless an exemption is approved by the MOE designated authority. An exemption would normally only be considered for the purposes of enhancing the quality of the winter range. An exemption is not required for harvesting of Western Red Cedar trees within UWR Unit No. UK1-09 by selective helicopter removal ("single-stem harvest"). Maximum opening size for this operation will be less than 0.1 hectare. Licensees will notify the regional office of Ministry of Environment prior to initiating operations such that monitoring to evaluate vegetation response and animal use may be considered.

   MOE recognizes that a 1 hectare helicopter drop zone is required for long term use in UWR Unit No. LKL-1. An exemption to general wildlife measure 3 will be considered subject to the extent of clearing and timing of use.

4. An exemption is not required for harvesting within the designated ungulate winter ranges when it is required to address worker safety: felling of danger trees, felling for guy line anchors, felling of tail hold anchor trees within an UWR along adjacent cutblock boundaries. Harvesting will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat within the designated ungulate winter ranges.

5. Trees that must be felled within an UWR will be left onsite to provide coarse woody debris, unless the felled tree lies outside the UWR.

6. Salvage harvesting is not to occur within the designated ungulate winter ranges unless an exemption is approved by the MOE designated authority.

7. An authorized person who carries out a primary forest activity within 100m of a designated ungulate winter range, as per Schedule A, must ensure that the adjacent designated ungulate winter range is not materially adversely affected by foreseeable impacts of their activities.

Signed this 17 day of March, 2009
Doug Konkin, Deputy Minister
Ministry of Environment

BRITISH COLUMBIA

ORDER – UNGULATE WINTER RANGE #U1-006

The following order applies to the area identified within the attached Schedule A and takes effect on the 18 day of October, 2004.

This order is given under the authority of sections 7(2) and 10(1) of the *Government Actions Regulation* (B.C. Reg. 17/04).

The Deputy Minister of Water, Land and Air Protection orders that:

1. the ungulate winter range shown in the map set out in the attached Schedule A (#U1-006) is established;
2. the ungulate winter range in the attached Schedule A is established for black-tailed deer (*Odocoileus hemionus columbianus*) and Roosevelt elk (*Cervus elaphus roosevelti*)*;* and
3. the general wildlife measures outlined in Schedule 1 are established for the ungulate winter range in the attached Schedule A:

Schedule 1 – General Wildlife Measures

1. Road construction is not to occur within the designated ungulate winter ranges unless there is no other practicable option and an exemption is approved by the MWLAP designated authority.

2. An exemption is not required for road maintenance, road deactivation, felling of danger trees or brushing and clearing on existing roads within the UWR. These activities will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat within the designated ungulate winter range.

3. Harvesting is not to occur within the designated ungulate winter ranges unless an exemption is approved by the MWLAP designated authority. An exemption would be considered for the purposes of enhancing the quality of the winter range.

4. An exemption is not required for harvesting within the designated ungulate winter ranges when it is required to address worker safety: felling of danger trees, felling for guy line anchors, felling of tail hold anchor trees within an UWR along adjacent cutblock boundaries. Harvesting will be conducted in a manner that does not result in a material adverse impact on the ungulate winter range habitat within the designated ungulate winter ranges.

5. Trees that must be felled within an UWR will be left onsite to provide coarse woody debris, unless the felled tree lies outside the UWR.

6. Salvage harvesting is not to occur within the designated ungulate winter ranges unless an exemption is approved by the MWLAP designated authority.

ORIGINAL SIGNED BY

_____

Signed this _____ day of _____, 2004
Gordon Macatee, Deputy Minister
Ministry of Water, Land and Air Protection

Appendix G Invasive Plants

FOR PURPOSES OF FRPA, THE PRESCRIBED SPECIES OF INVASIVE PLANTS ARE AS FOLLOWS (S.2 INVASIVE PLANTS REGULATION, 2004):

Weed Species	**Scientific name**	**Weed Species**	**Scientific name**
Anchusa	Anchusa officinalis	Oxeye daisy	Chrysanthemum leucanthemem
Baby's breath	Gypsophila paniculata	Perennial pepperweed	Lepidium latifolium
Black knapweed	Centaurea nigra	Plumeless thistle	Carduus acanthoides
Blueweed	Echium vulgare	Puncture vine	Tribulus terrestris
Brown knapweed	Centaurea jacea	Purple loosestrife	Lythrum salicaria
Bull thistle	Cirsium vulgare	Rush skeletonweed	Chondrilla juncea
Canada thistle	Cirsium arvense	Russian knapweed	Acroptilon repens
Common burdock	Arctium minus	Scentless chamomile	Matricaria maritima
Common tansy	Tanacetum vulgare	Scotch broom	Cytisus scoparius
Dalmatian toadflax	Linaria dalmatica	Scotch thistle	Onopordum acanthium
Diffuse knapweed	Centaurea diffusa	Spotted knapweed	Centaurea maculosa
Field scabious	Knautia arvensis	St. John's wort	Hypericum perforatum
Giant knotweed	Polygonum sachalinense	Sulphur cinquefoil	Potentilla recta
Gorse	Ulex europaeus	Tansy ragwort	Senecio jacobaea
Hoary alyssum	Berteroa incana	Teasel	Dipsacus fullonum
Hoary cress	Cardaria draba	Yellow Iris	Iris pseudacorus
Hound's-tongue	Cynoglossum officinale	Yellow starthistle	Centaurea solstitialis
Japanese knotweed	Polygonum cuspidatum	Yellow toadflax	Linaria vulgaris
Leafy spurge	Euphorbia esula		
Marsh thistle	Cirsium palustre		
Meadow hawkweed	Hieracium pilosella.		
Meadow knapweed	Centaurea pratensis		
Nodding thistle	Carduus nutans		
Orange hawkweed	Hieracium aurantiacum		

Coastal ISC Priority Invasive Plant List (current to: April 21, 2016)

PREVENT

These species are not known to occur in the region, but are likely to establish if introduced.
Eradicate if found. REPORT ALL SIGHTINGS

Plant Species	Status	Report To
Common Crupina *Crupina vulgaris***	**Provincial EDRR**	Report A Weed BC
Cordgrass, Smooth *Spartina alterniflora***	**Provincial EDRR**	Report A Weed BC
Hawkweed, Whiplash *Hieracium flagellare*	Regional EDRR	Info@coastalisc.com
Knapweed, Russian *Acroptilon repens*	Regional EDRR	Info@coastalisc.com
Kudzu *Pueraria Montana***	**Provincial EDRR**	Report A Weed BC
Rush Skeleton weed *Chondrilla juncea*	Regional EDRR	Info@coastalisc.com
Yellow Starthistle *Centaurea solstitialis***	**Provincial EDRR**	Report A Weed BC

ERADICATE

These species are known to occur in limited distribution and low density.
Eradicate if found. REPORT ALL SIGHTINGS

Plant Species	Status	Report To
Blueweed, *Echium vulgare*	Regional EDRR	Info@coastalisc.com
Buffalo Burr, *Solanum rostratum*	Regional EDRR	Info@coastalisc.com
Common Reed, *Phragmites australis***	**Provincial EDRR**	Report A Weed BC
Cordgrass, Dense-flowered *Spartina densiflora***	**Provincial EDRR**	Report A Weed BC
Cordgrass, English *Spartina anglica***	**Provincial EDRR**	Report A Weed BC
Cordgrass, Salt meadow *Spartina patens***	**Provincial EDRR**	Report A Weed BC
Garlic Mustard *Alliaria petiolata*	Regional EDRR	Info@coastalisc.com
Giant Hogweed *Heracleum mantegazzianum* (T) (N)	Regional EDRR	Info@coastalisc.com
Giant Reed *Arundo donax***	**Provincial EDRR**	Report A Weed BC
Hoary Alyssum *Berteroa incana*	Regional EDRR	Info@coastalisc.com
Hoary Cress, Heart-pod *Lepidium draba subsp. Draba*	Regional EDRR	Info@coastalisc.com
Lesser Celendine, *Ficaria verna*	Regional EDRR	Info@coastalisc.com
Loosestrife, Garden (Yellow) *Lysimachia vulgaris*	Regional EDRR	Info@coastalisc.com
Milk Thistle *Silybum marianum* (N)	Regional EDRR	Info@coastalisc.com
Shiney Geranium, *Geranium lucidum***	**Provincial EDRR**	Report A Weed BC
Slender False Brome, *Brachypodium sylvaticum***	**Provincial EDRR**	Report A Weed BC
Sulfur cinquefoil *Potentilla recta*	Regional EDRR	Info@coastalisc.com
Sweet Fennel *Foeniculum vulgare*	Regional EDRR	Info@coastalisc.com
Wild Chervil *Anthriscus sylvestris*	Regional EDRR	Info@coastalisc.com

CONTAIN

These species have established infestation in portions of the region.
Contain existing infestation and prevent spread to un-infested areas.

Plant Species

Carpet Burweed *Soliva sessilis*
Hawkweed, Orange *Hieracium aurantiacum*
Knapweed, Black *Centaurea nigra*
Knapweed, Diffuse *Centaurea diffusa* (N)
Knapweed, Meadow *Centaurea pratensis*
Knapweed, Spotted *Centaurea maculosa* (B) (N)
Knotweed, Bohemian *Fallopia x bohemica* (N)
Knotweed, Giant *Fallopia sachalinensis* (N)
Knotweed, Himalayan *Polygonum polystachum* (N)
Knotweed, Japanese *Fallopia japonica* (N)
Poison Hemlock *Conium maculatum* (T)
Policemans Helmet/Himalayan Balsam *Impatiens glandulifera*
Scotch Thistle *Onopordum acanthium*
Yellow Flag Iris *Iris pseudacorus* (N)

CONTROL

Established infestations common and widespread throughout the Coastal ISC region.
Focus control in high value conservation areas.
Use biological control, if available, on a landscape scale.

Plant Species

Bur Chervil *Anthriscus caucalis* (N)
Burdock Species *Arctium spp.*
Canada Thistle *Cirsium arvense* (B) (N)
Tansy, Common *Tanacetum vulgare*
Teasel, Fuller's Dipsacus fullonum
Dalmatian Toadflax *Linaria dalmaticab* (B) (N)
English Holly *Ilex aquifolium*
English Ivy *Hedera helix*
Giant Mannagrass *Glyceria maxima*
Hairy Cat's Ear *Hypochaeris radicata*
Himalayan Blackberry *Rubus armeniacus* (discolor)
Jimsonweed/Devil's Apple *Datura stramonium* (T)
Periwinkle Species *Vinca spp.*
Loosestrife, Purple *Lythrum salicaria* (B) (N)
Scotch Broom *Cytisus scoparius*
St. John's Wort *Hypericum perforatum* (B)
Tansy Ragwort *Senecio jacobaea* (B) (N)

CONTROL

Established infestations common and widespread throughout the Coastal ISC region.
Focus control in high value conservation areas.
Use biological control, if available, on a landscape scale.

Plant Species

Tansy Ragwort *Senecio jacobaea* (B) (N)

Butterfly Bush *Buddleja davidii*
Daphne/Spurge-Laurel *Daphne laureola* (T)
Gorse *Ulex europaeus*
Eurasian Water-milfoil *Myriophyllum spicatum*
Yellow Archangel *Lamiastrum galiobdolon*
Hawkweed,Yellow *Hieracium caespitosum*

Supplemental Notes:

• The above lists has been approved by the Coastal ISC Board and developed in consultation with key land managers in the Coastal ISC service area a the annual operational planning meeting (Februrary 2016).

• The above lists reflect the entire Coastal ISC area. The placement of a species into a category at the landscape level is very likely to be different from a placement of a species into a category at the local level.

• Provincial EDRR - provincially significant and are to be reported immediately to the province through Report-A-Weed.

• Regional EDRR - regionally significant species and to be reported to the Coastal ISC.

** BC Proposed Prohibited Weeds (PDF, February 2015)

(B) = Invasive plants with biological control agents available

(T) = Invasive plants which pose potential human health and safety hazards

(N) = BC Weed Control Act, Regulated Noxious Weed in BC

Appendix H Referrals and Correspondence

The information contained in this Appendix is considered Confidential and will only be distributed to the Ministry of Forests, Range, Natural Resource Operations and Rural Development for purposes of meeting Division 3 – Public Review and Comment of the *FPPR*.